

BFS Traversal

Step 1: Start at E

Legend

Complete

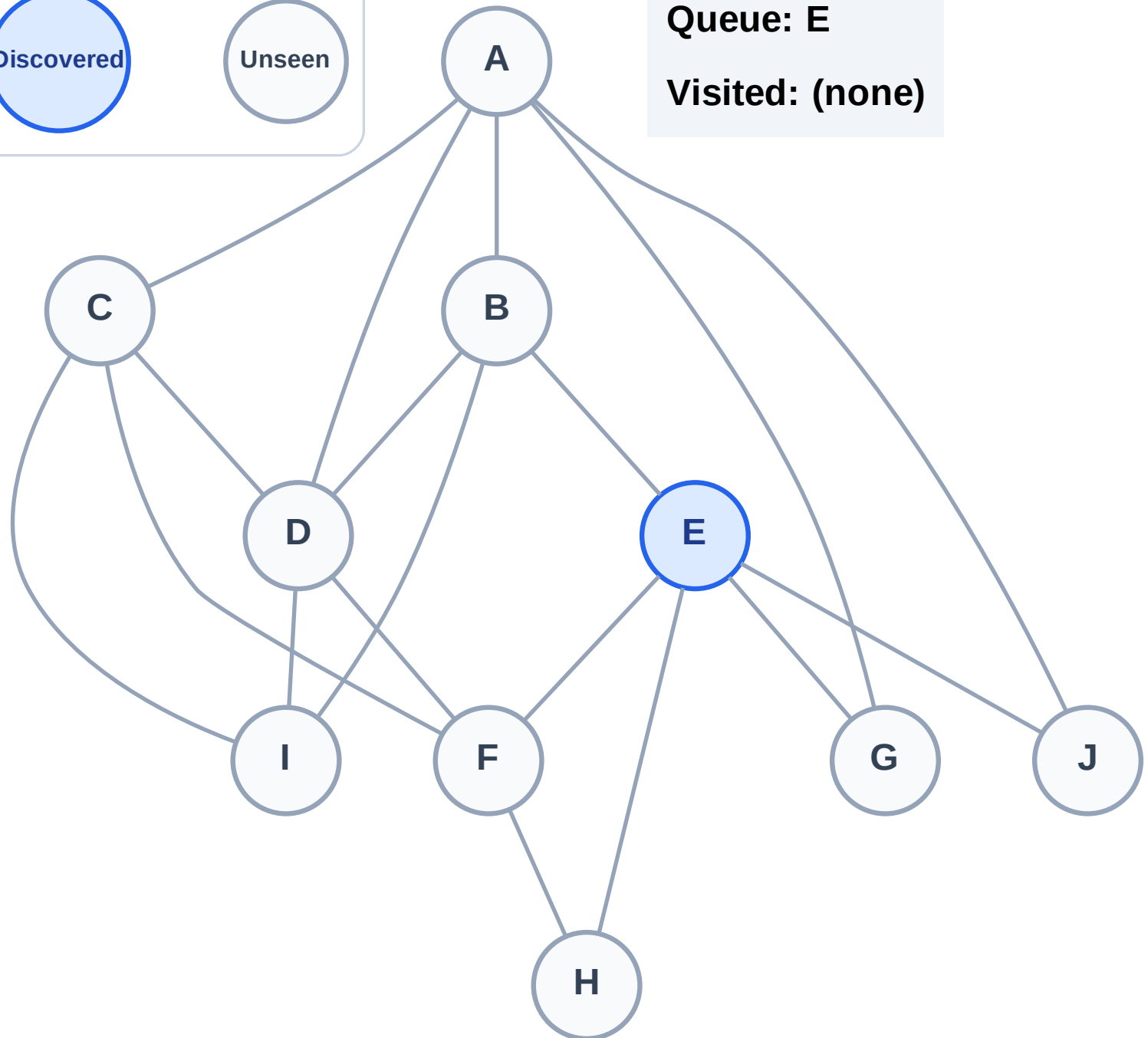
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

Complete

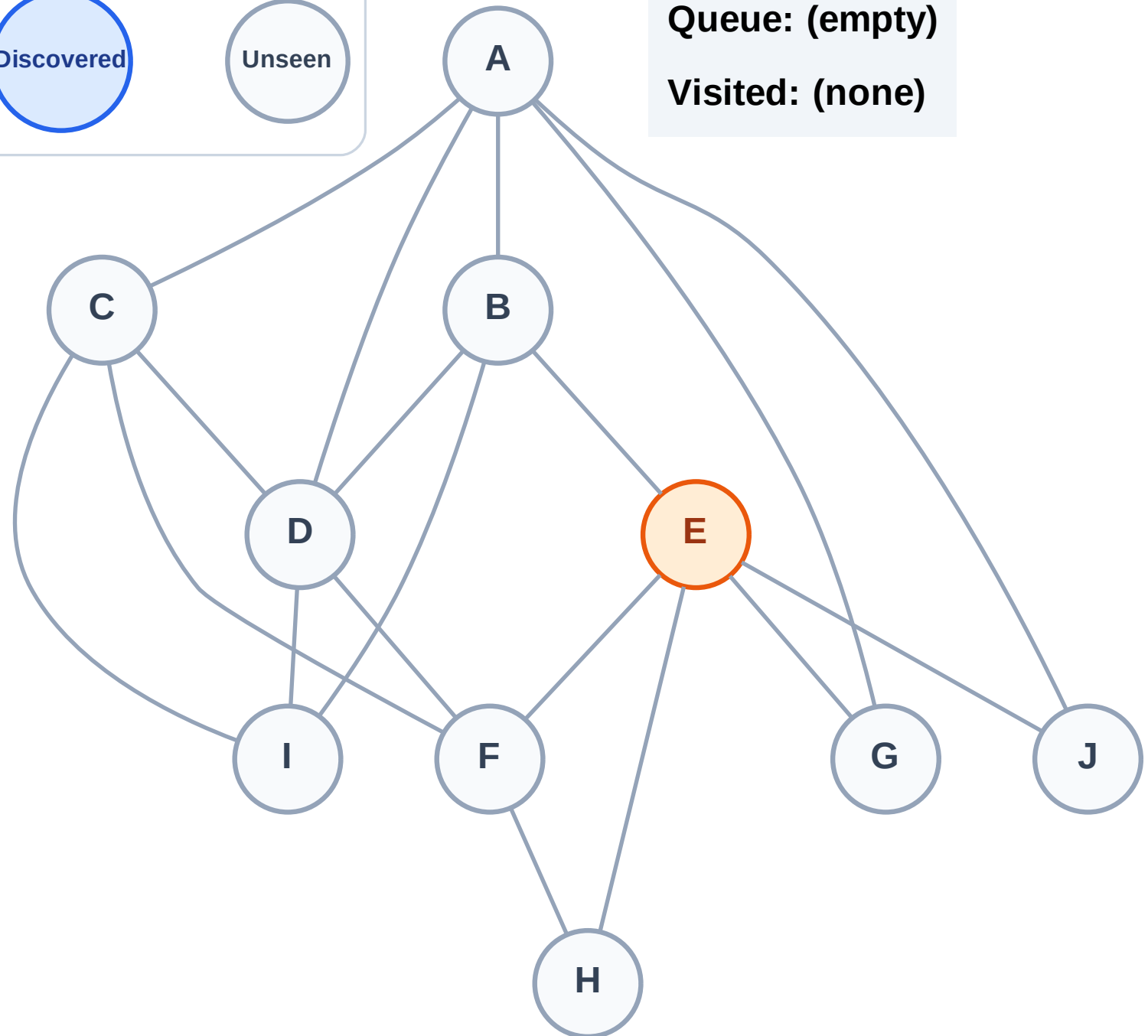
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, F, G, H, J from E

Legend

Complete

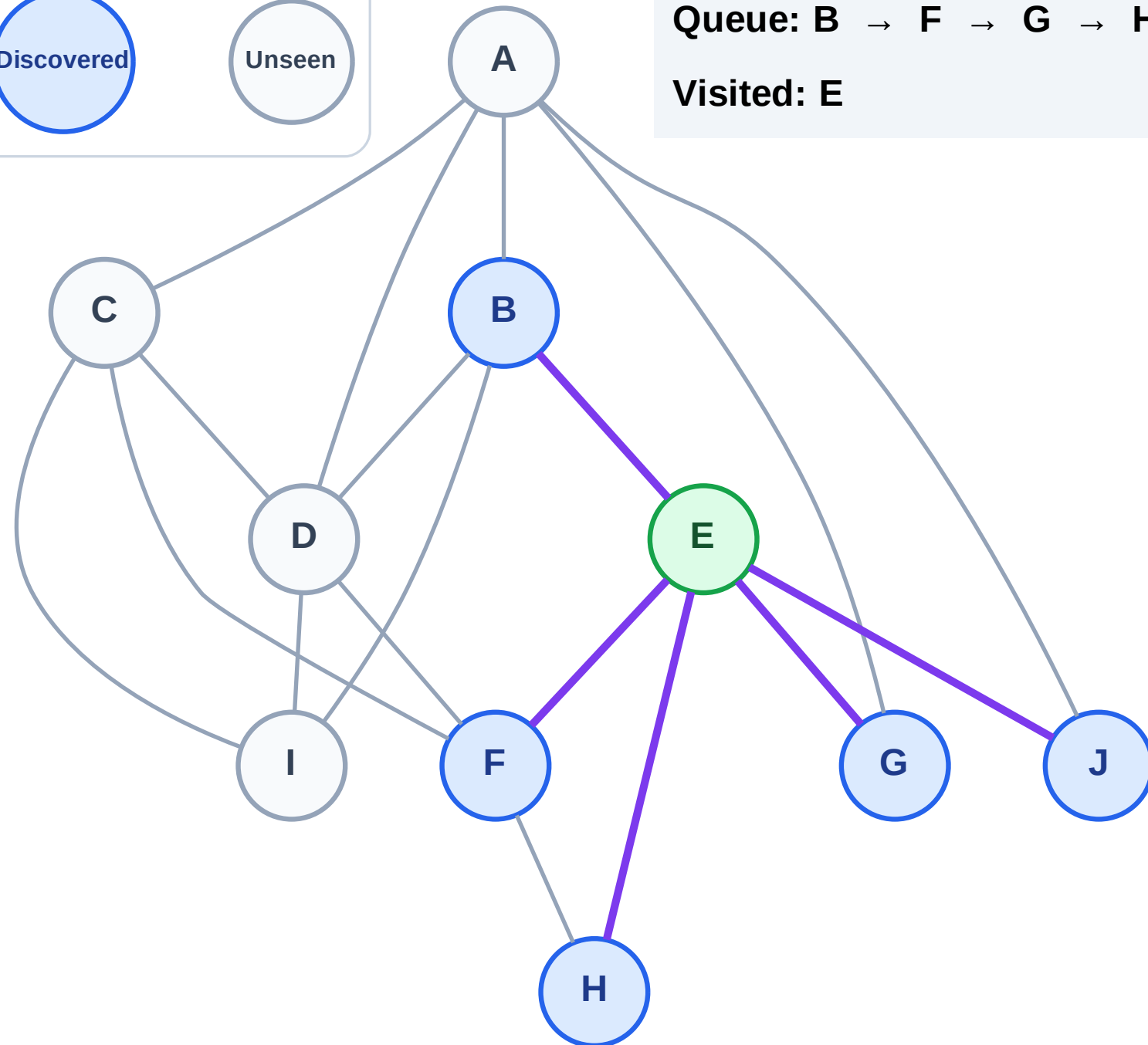
Processing

Discovered

Unseen

Queue: B → F → G → H → J

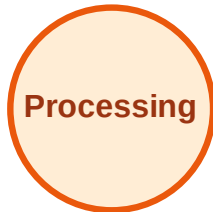
Visited: E



BFS Traversal

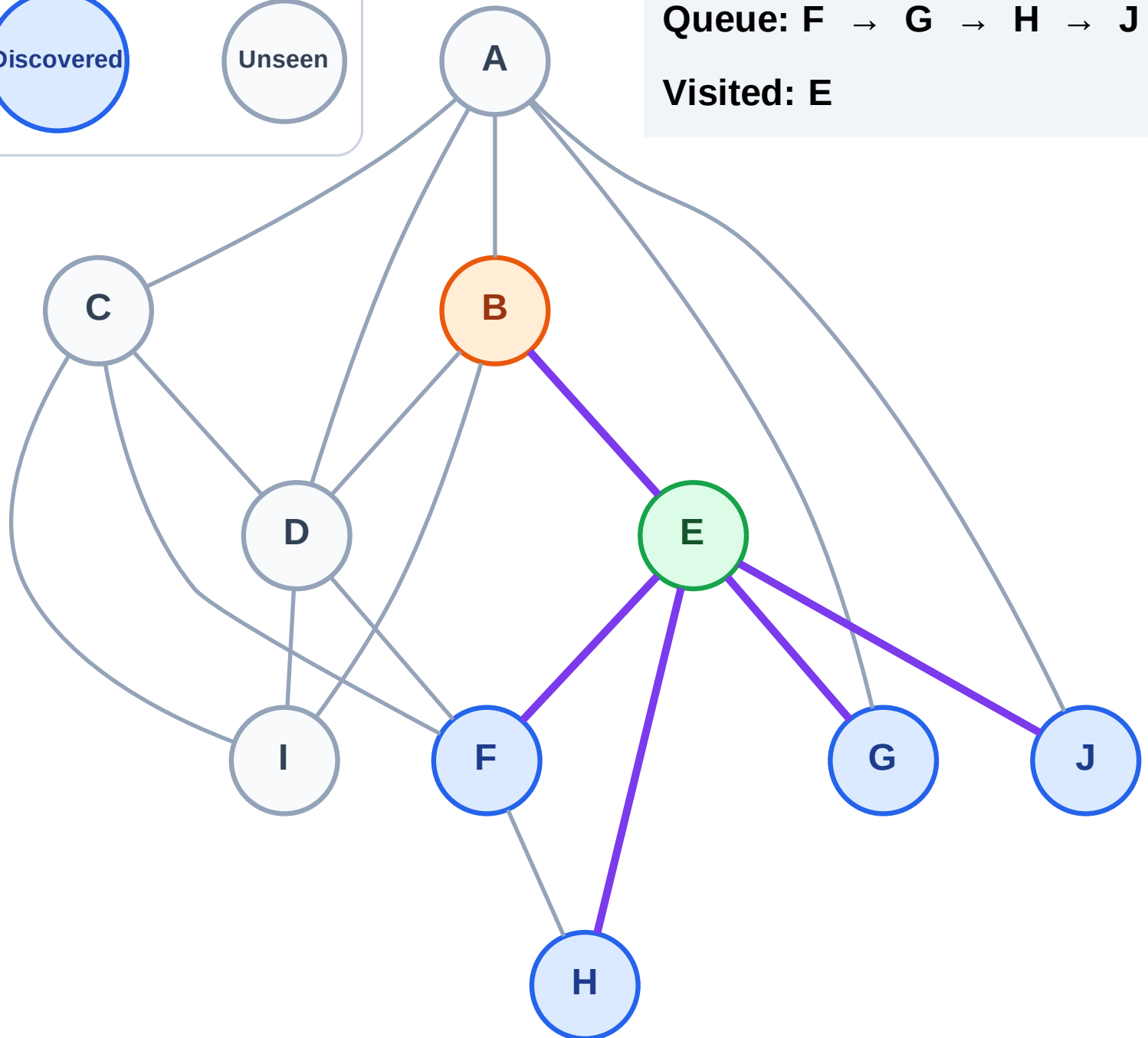
Step 4: Process node B

Legend



Queue: F → G → H → J

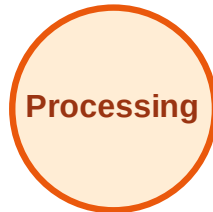
Visited: E



BFS Traversal

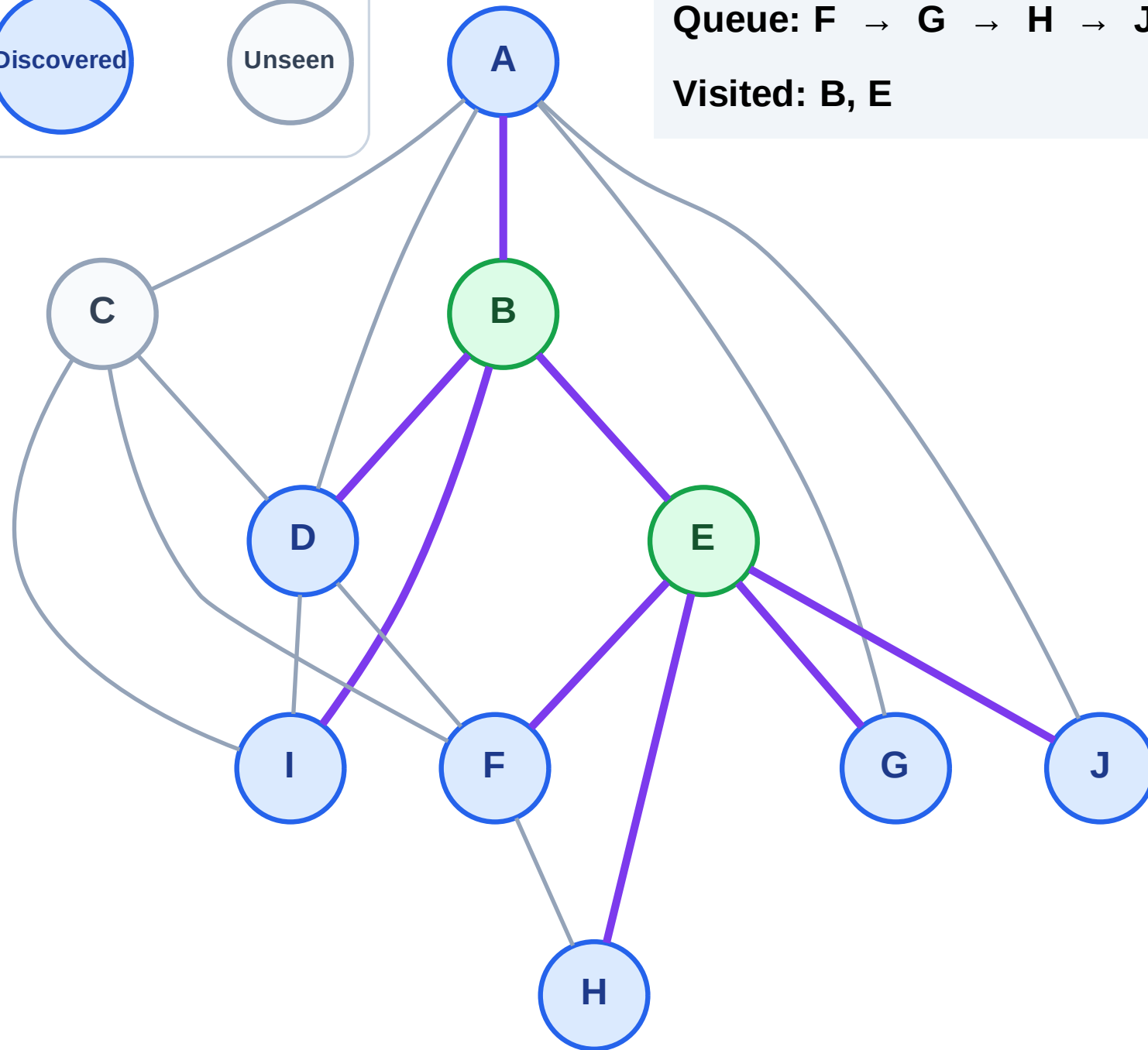
Step 5: Discover A, D, I from B

Legend



Queue: F → G → H → J → A → D → I

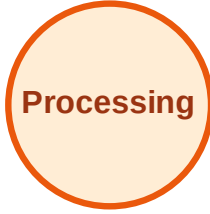
Visited: B, E



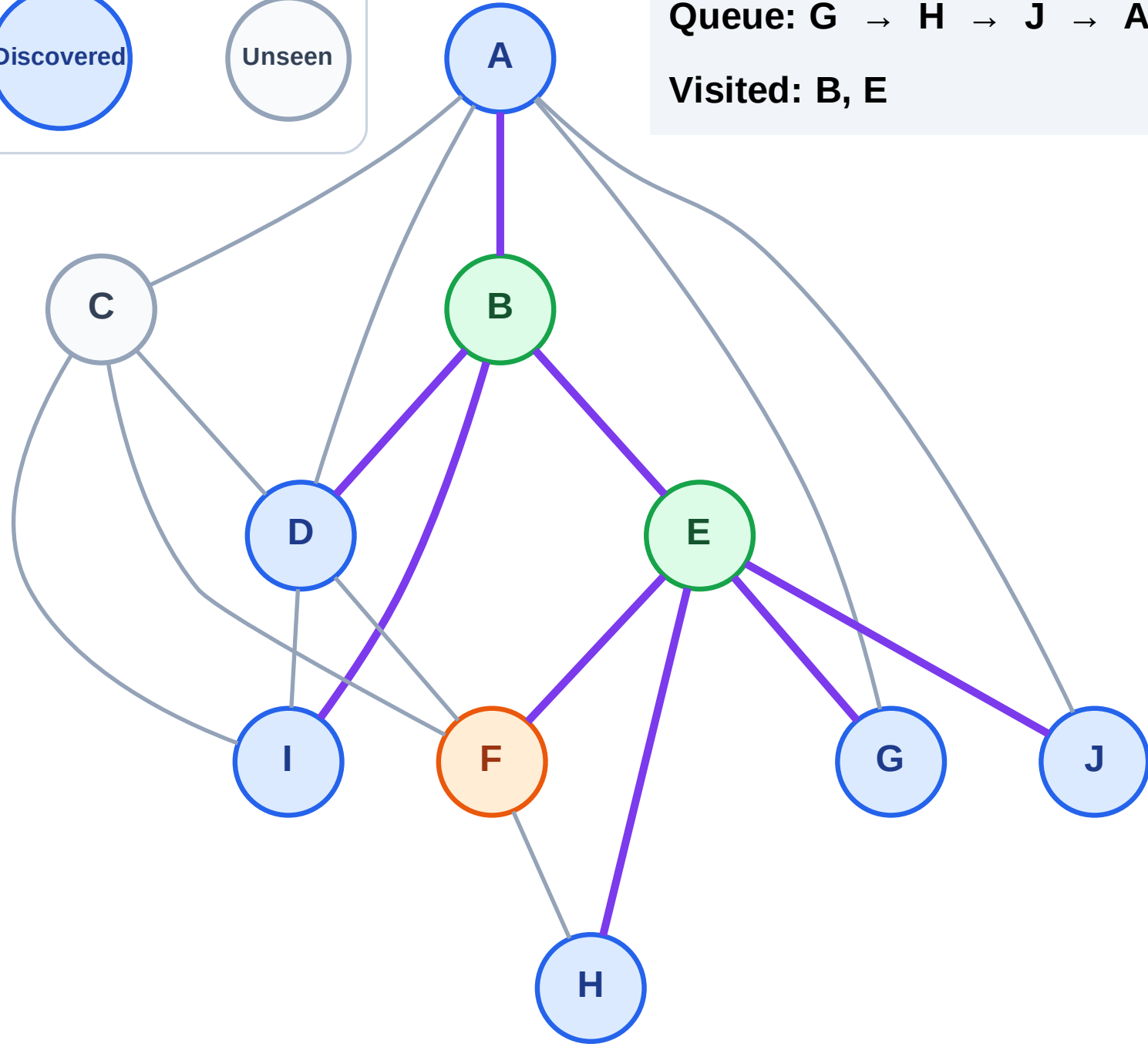
BFS Traversal

Step 6: Process node F

Legend



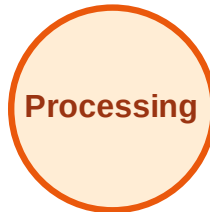
Queue: G → H → J → A → D → I
Visited: B, E



BFS Traversal

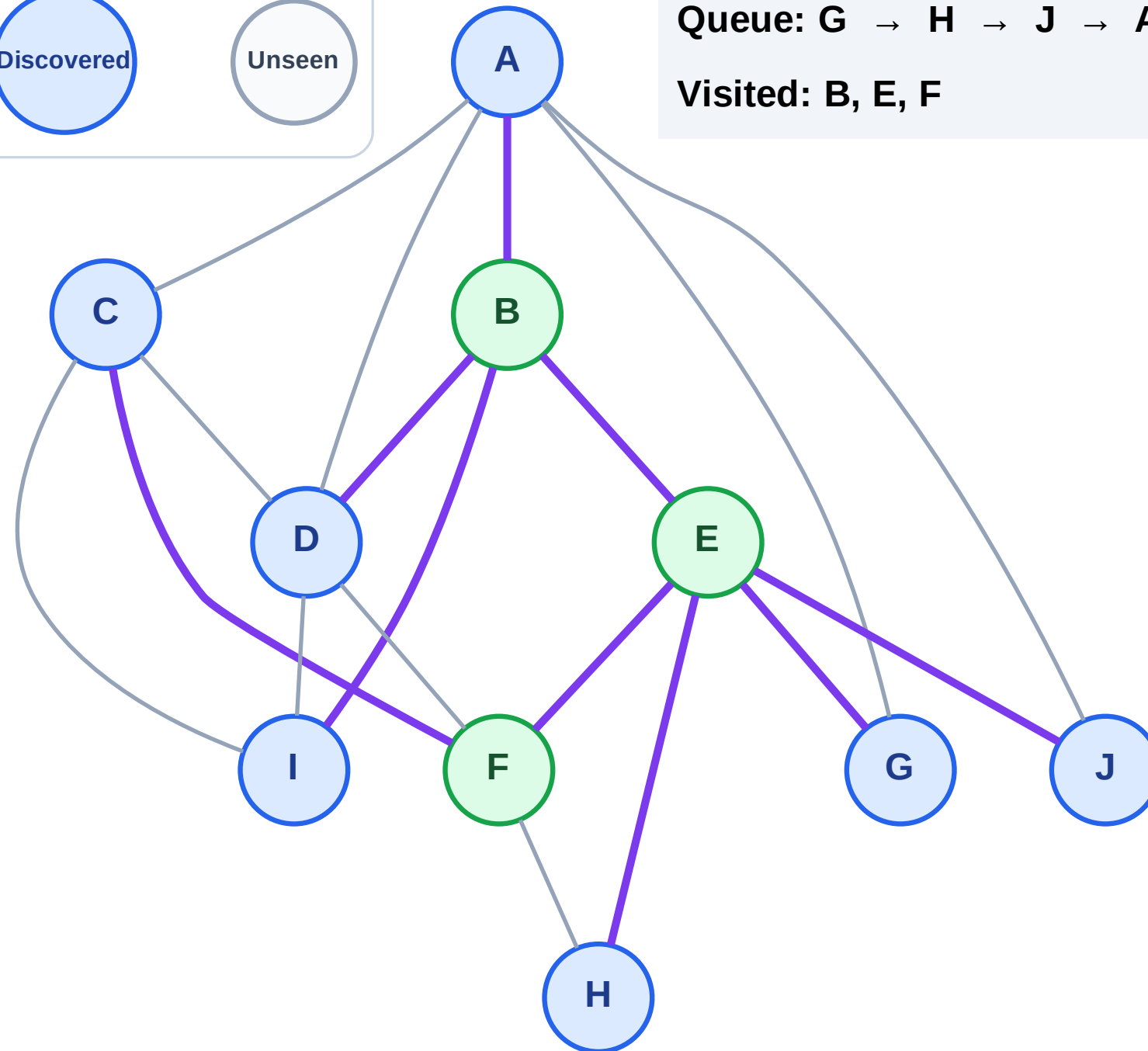
Step 7: Discover C from F

Legend



Queue: G → H → J → A → D → I → C

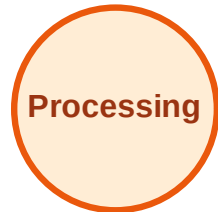
Visited: B, E, F



BFS Traversal

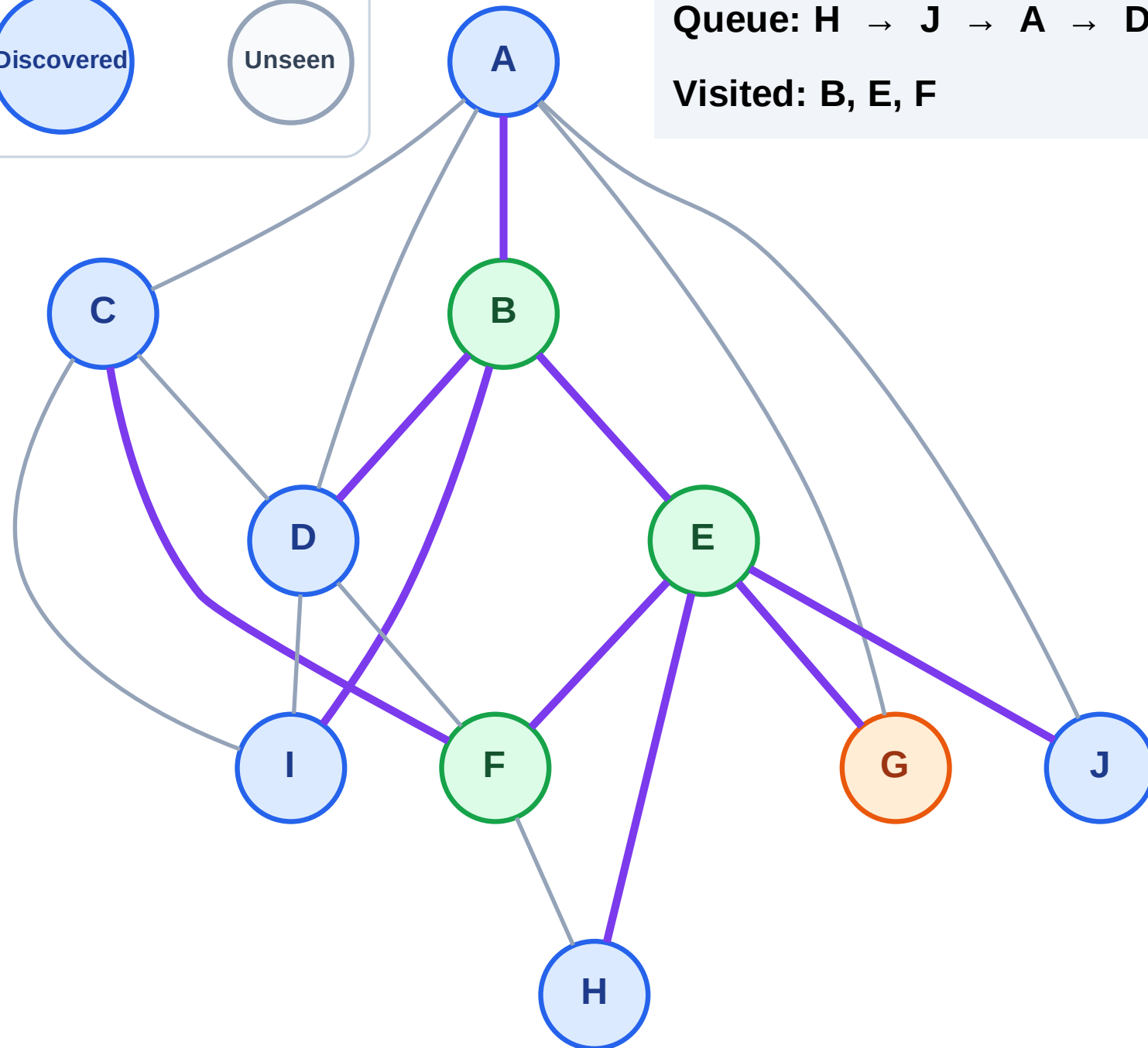
Step 8: Process node G

Legend



Queue: H → J → A → D → I → C

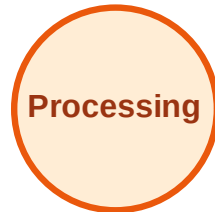
Visited: B, E, F



BFS Traversal

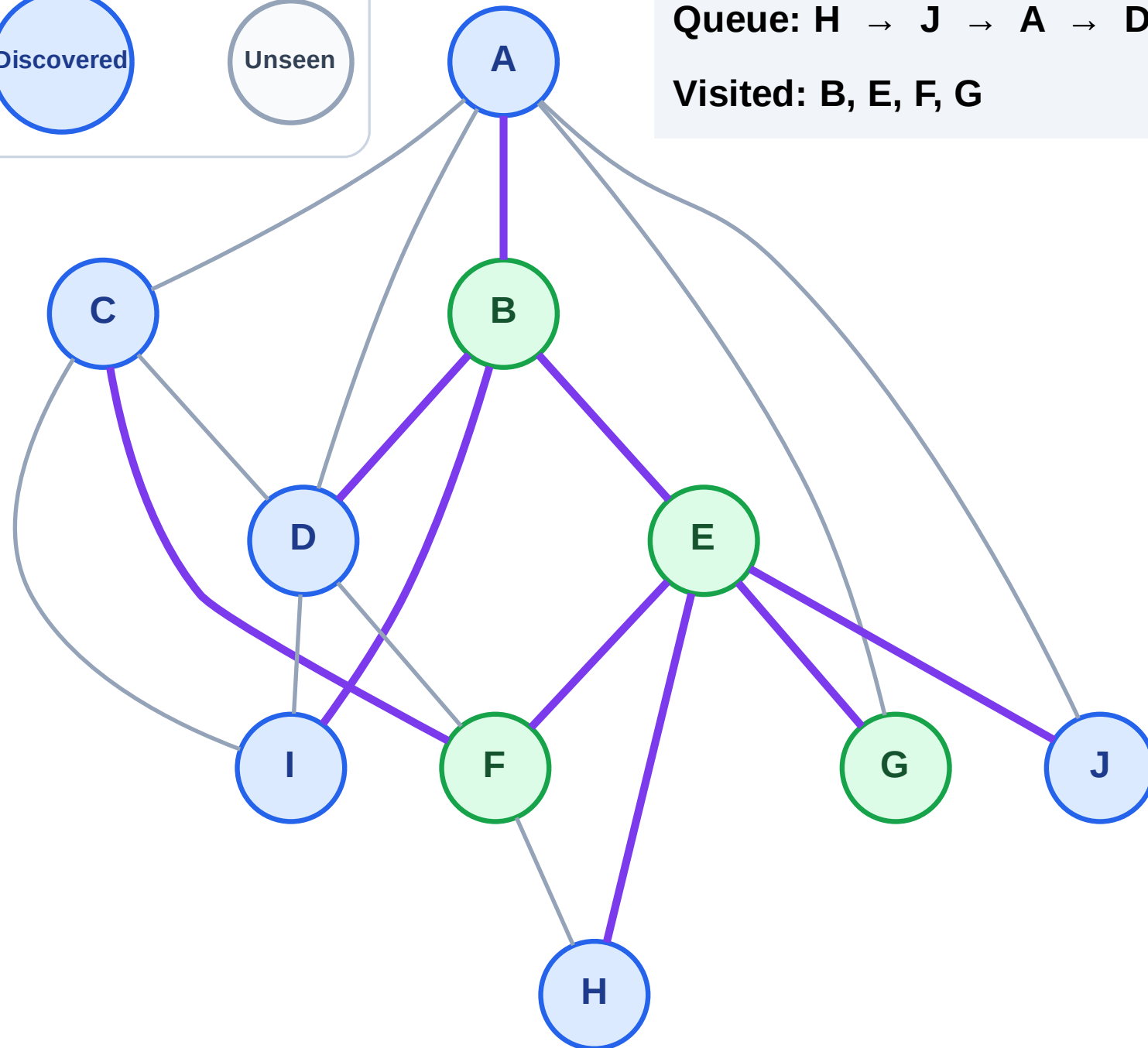
Step 9: No new neighbors from G

Legend



Queue: H → J → A → D → I → C

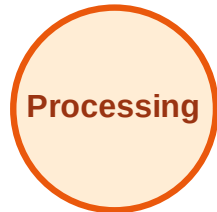
Visited: B, E, F, G



BFS Traversal

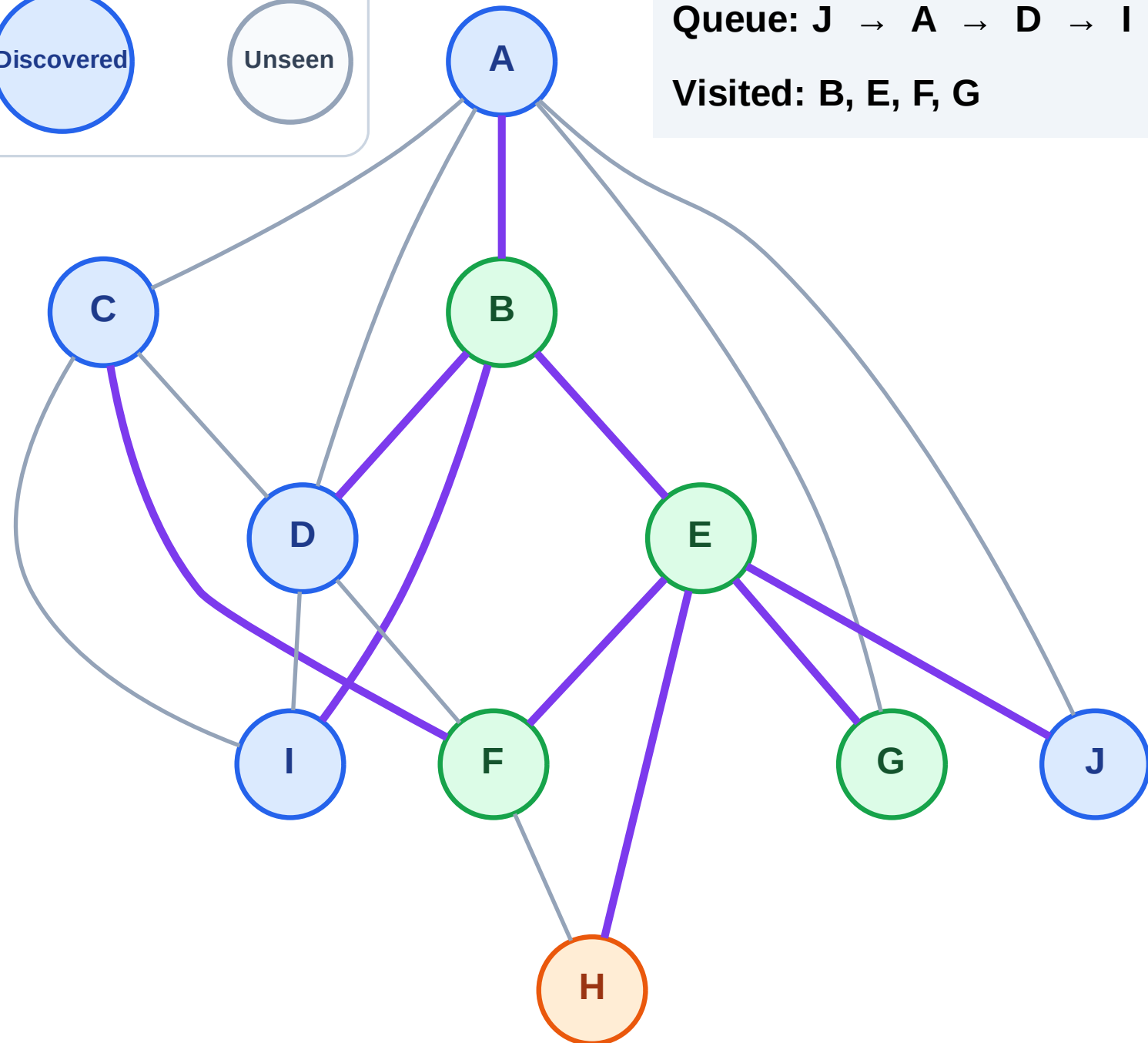
Step 10: Process node H

Legend



Queue: J → A → D → I → C

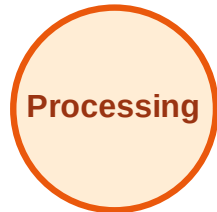
Visited: B, E, F, G



BFS Traversal

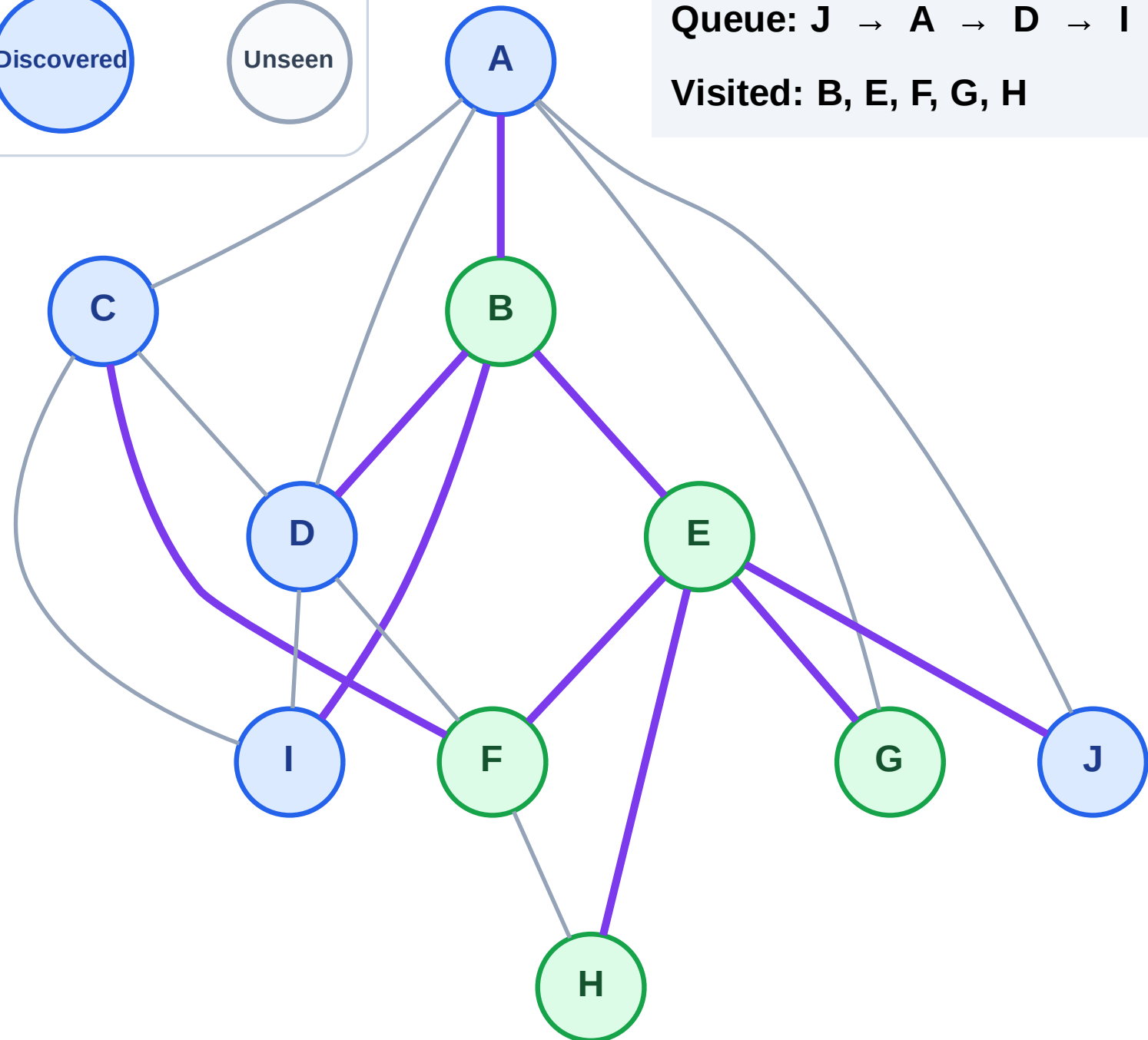
Step 11: No new neighbors from H

Legend



Queue: J → A → D → I → C

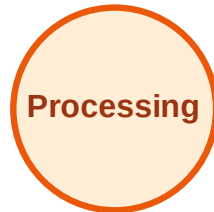
Visited: B, E, F, G, H



BFS Traversal

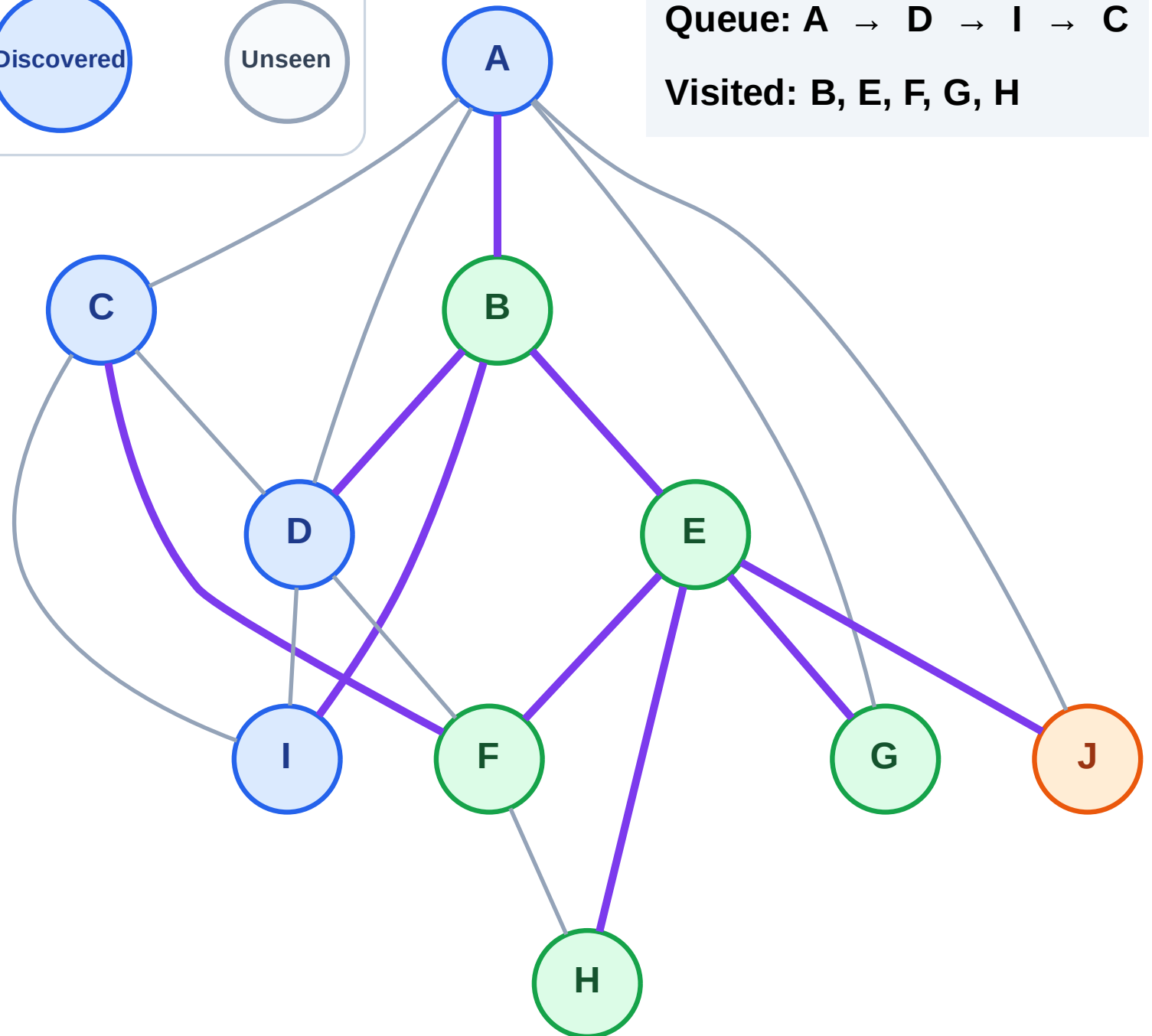
Step 12: Process node J

Legend



Queue: A → D → I → C

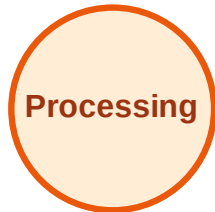
Visited: B, E, F, G, H



BFS Traversal

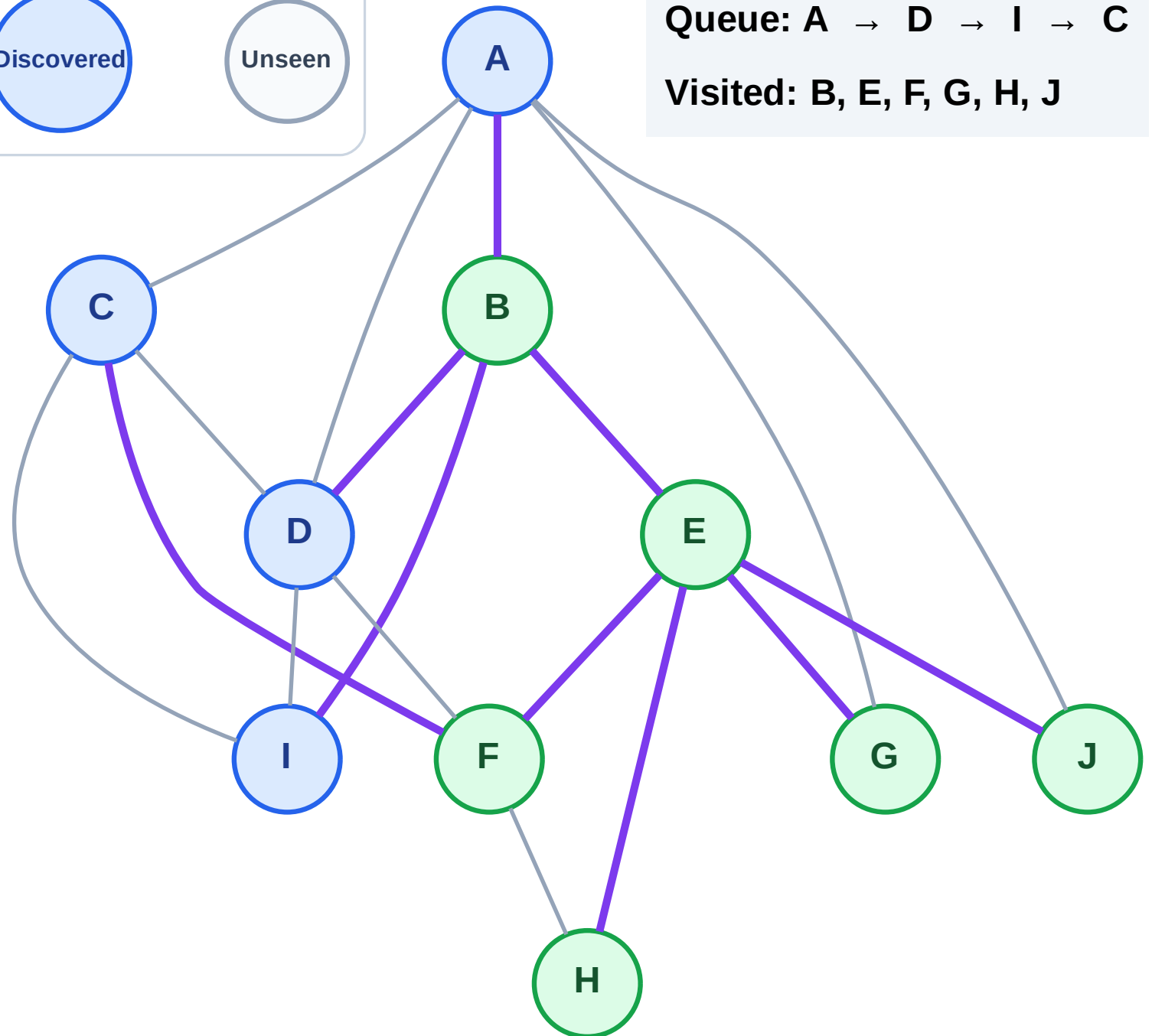
Step 13: No new neighbors from J

Legend



Queue: A → D → I → C

Visited: B, E, F, G, H, J



BFS Traversal

Step 14: Process node A

Legend

Complete

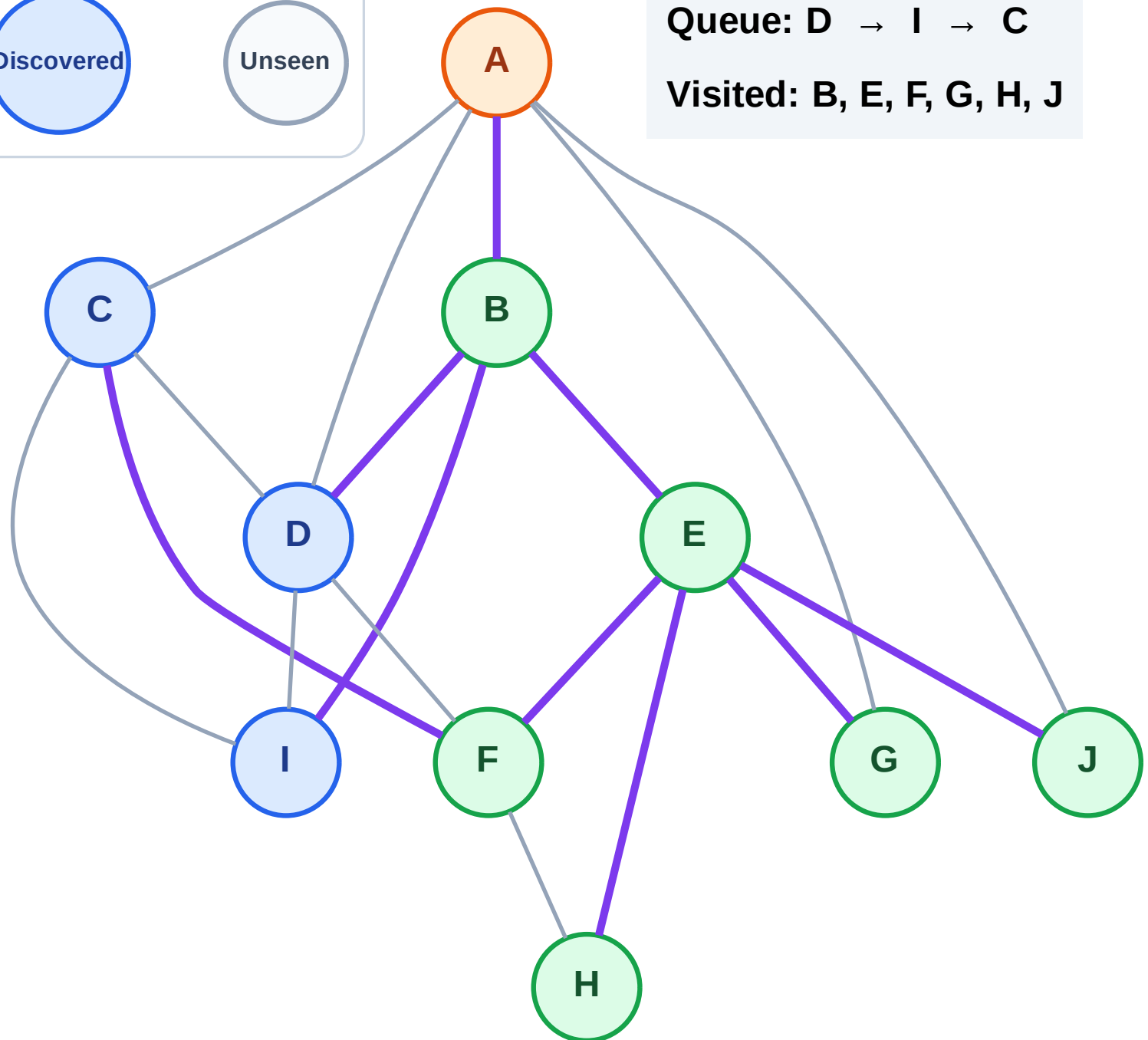
Processing

Discovered

Unseen

Queue: D → I → C

Visited: B, E, F, G, H, J



BFS Traversal

Step 15: No new neighbors from A

Legend

Complete

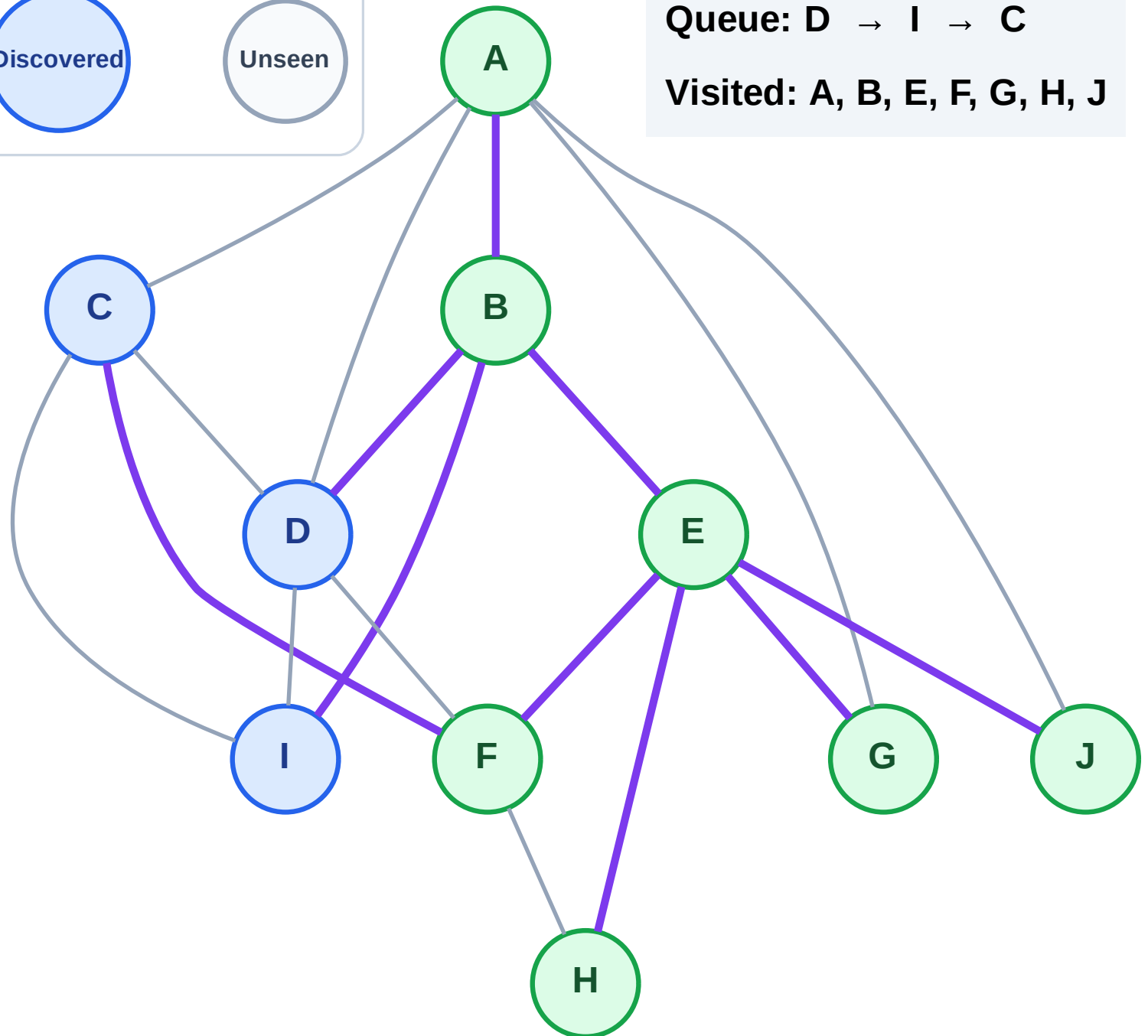
Processing

Discovered

Unseen

Queue: D → I → C

Visited: A, B, E, F, G, H, J



BFS Traversal

Step 16: Process node D

Legend

Complete

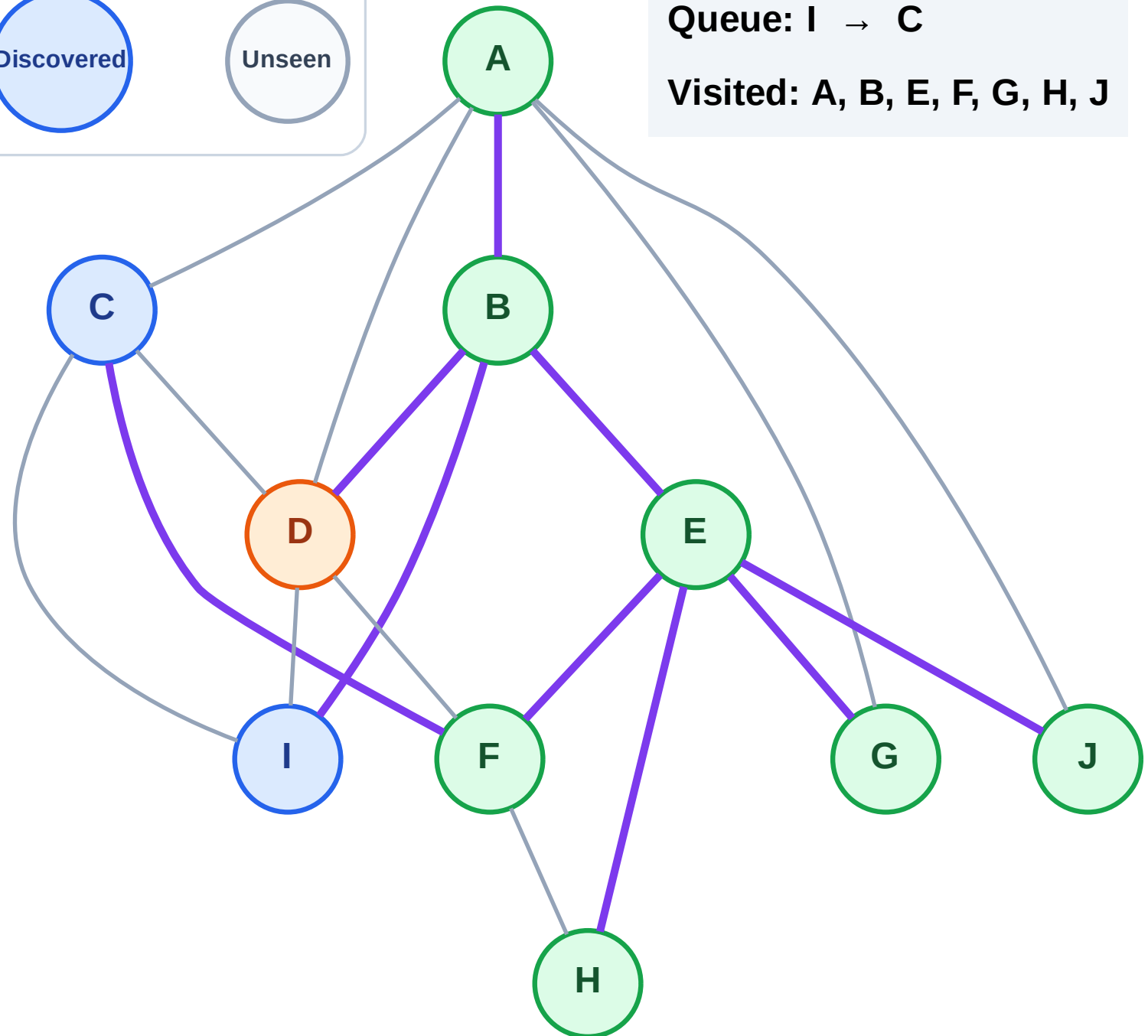
Processing

Discovered

Unseen

Queue: I → C

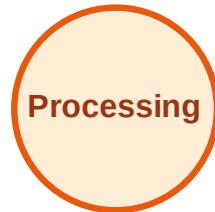
Visited: A, B, E, F, G, H, J



BFS Traversal

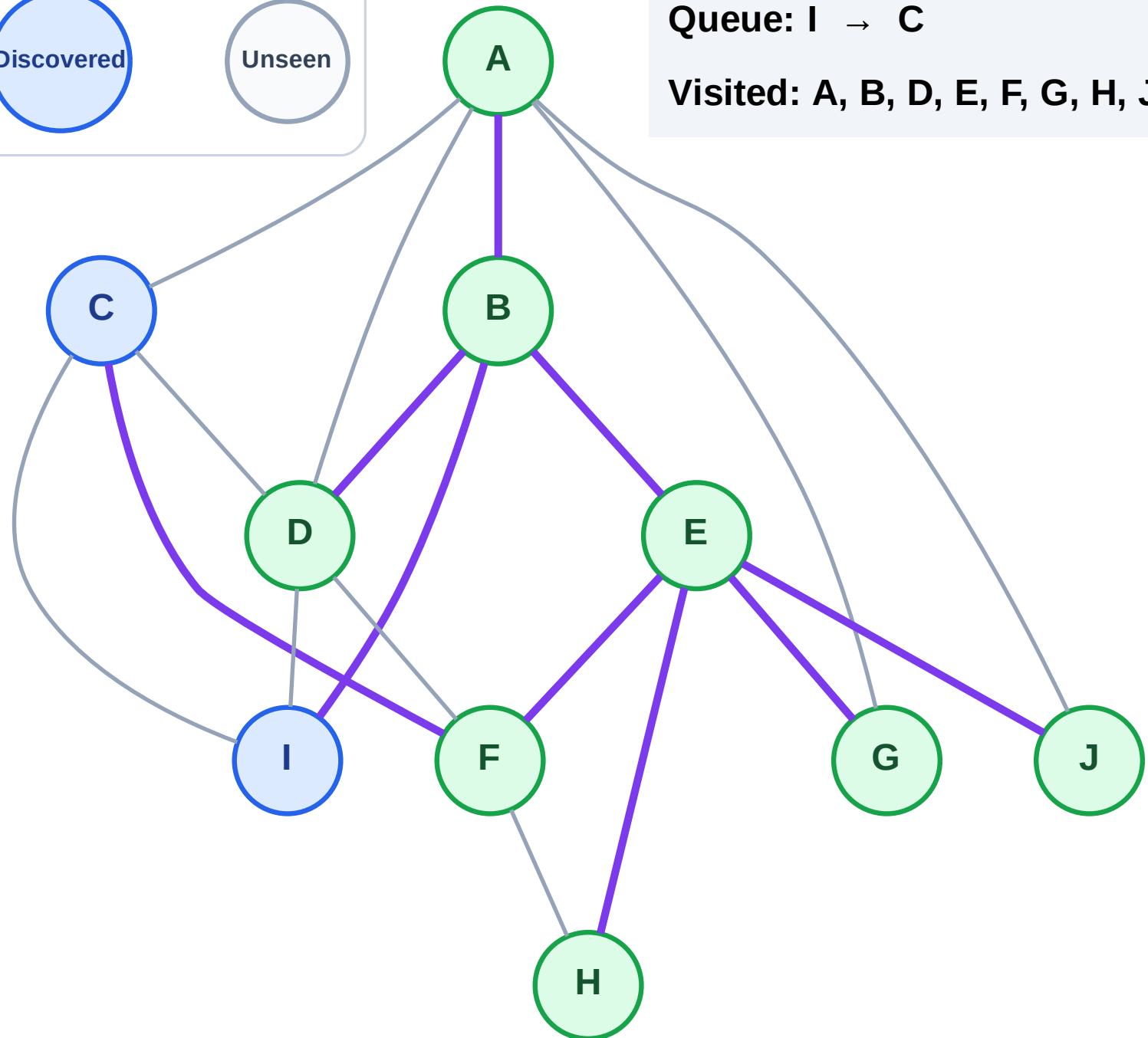
Step 17: No new neighbors from D

Legend



Queue: I → C

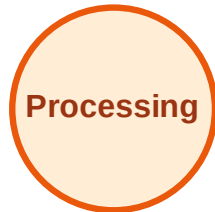
Visited: A, B, D, E, F, G, H, J



BFS Traversal

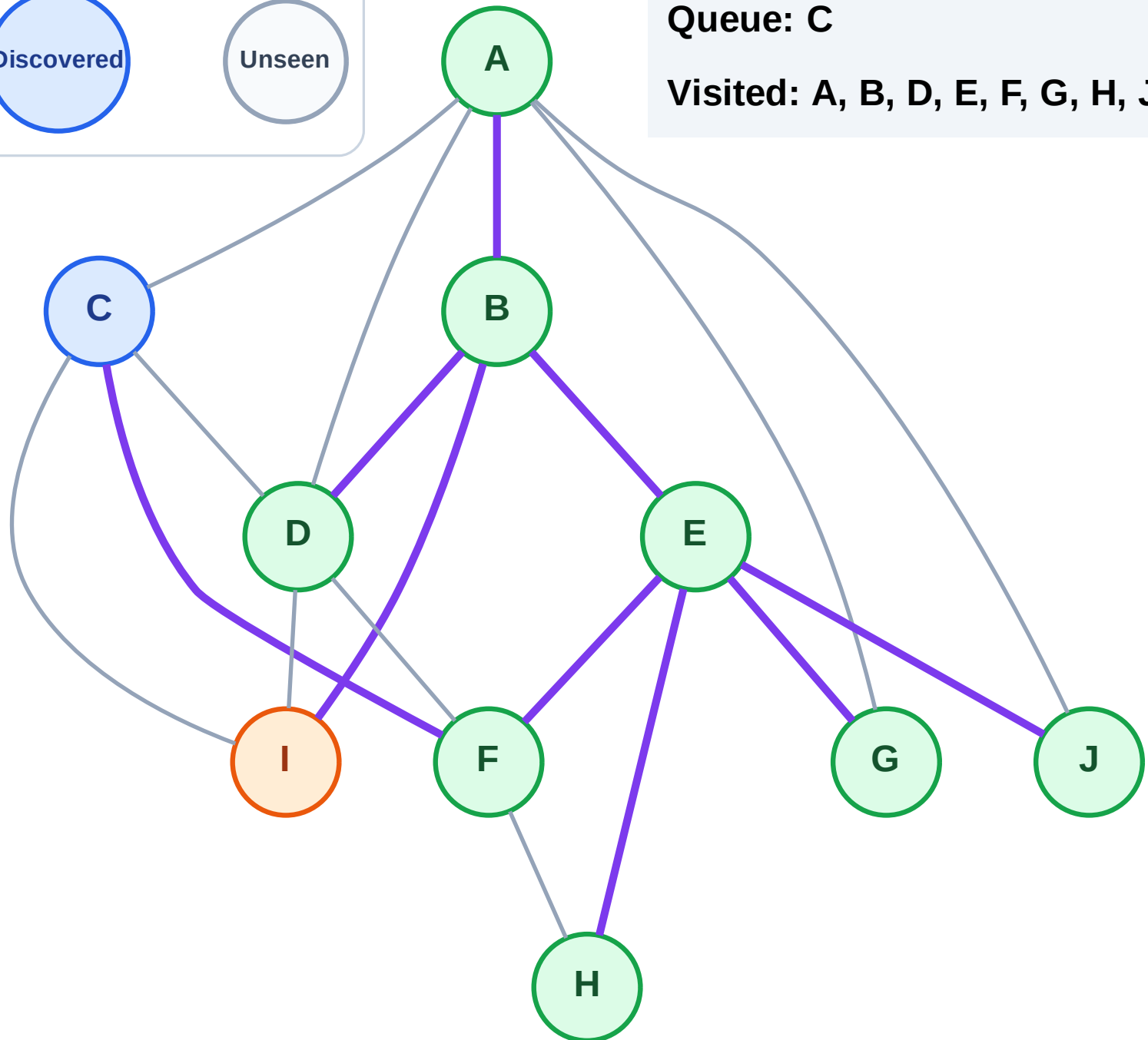
Step 18: Process node I

Legend



Queue: C

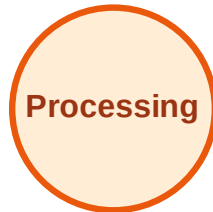
Visited: A, B, D, E, F, G, H, J



BFS Traversal

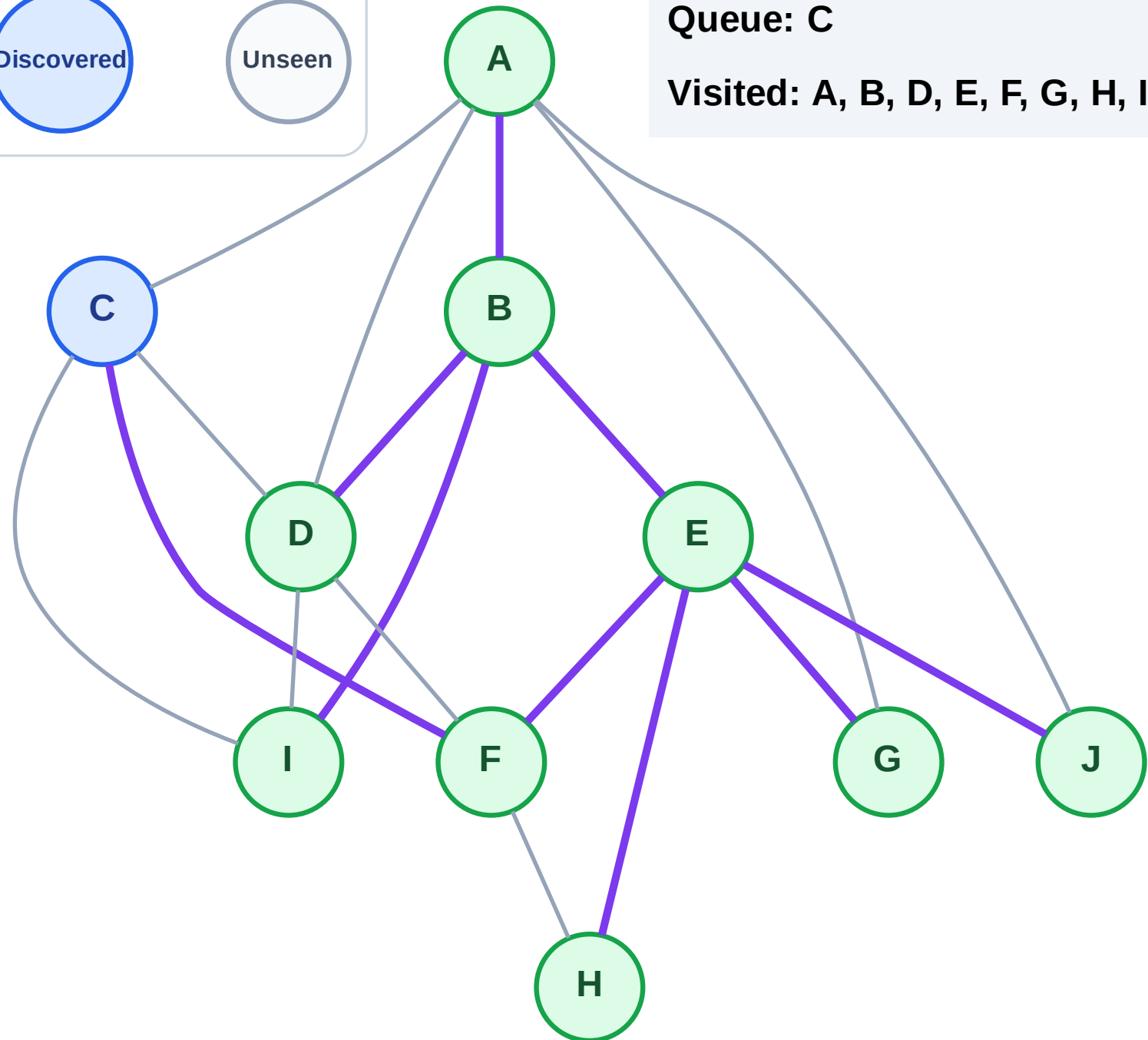
Step 19: No new neighbors from I

Legend



Queue: C

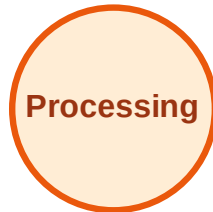
Visited: A, B, D, E, F, G, H, I, J



BFS Traversal

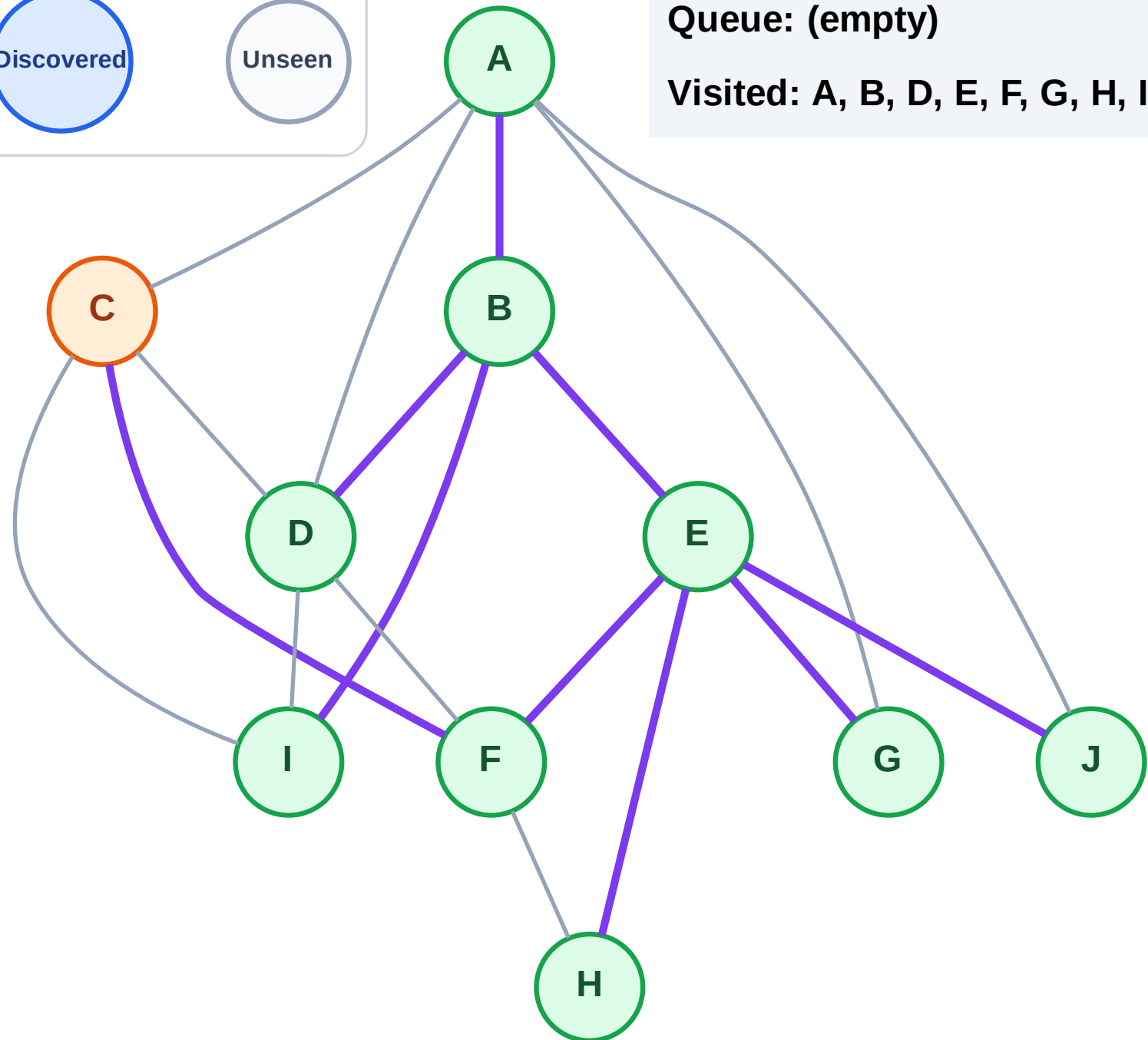
Step 20: Process node C

Legend



Queue: (empty)

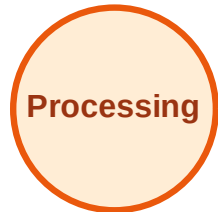
Visited: A, B, D, E, F, G, H, I, J



BFS Traversal

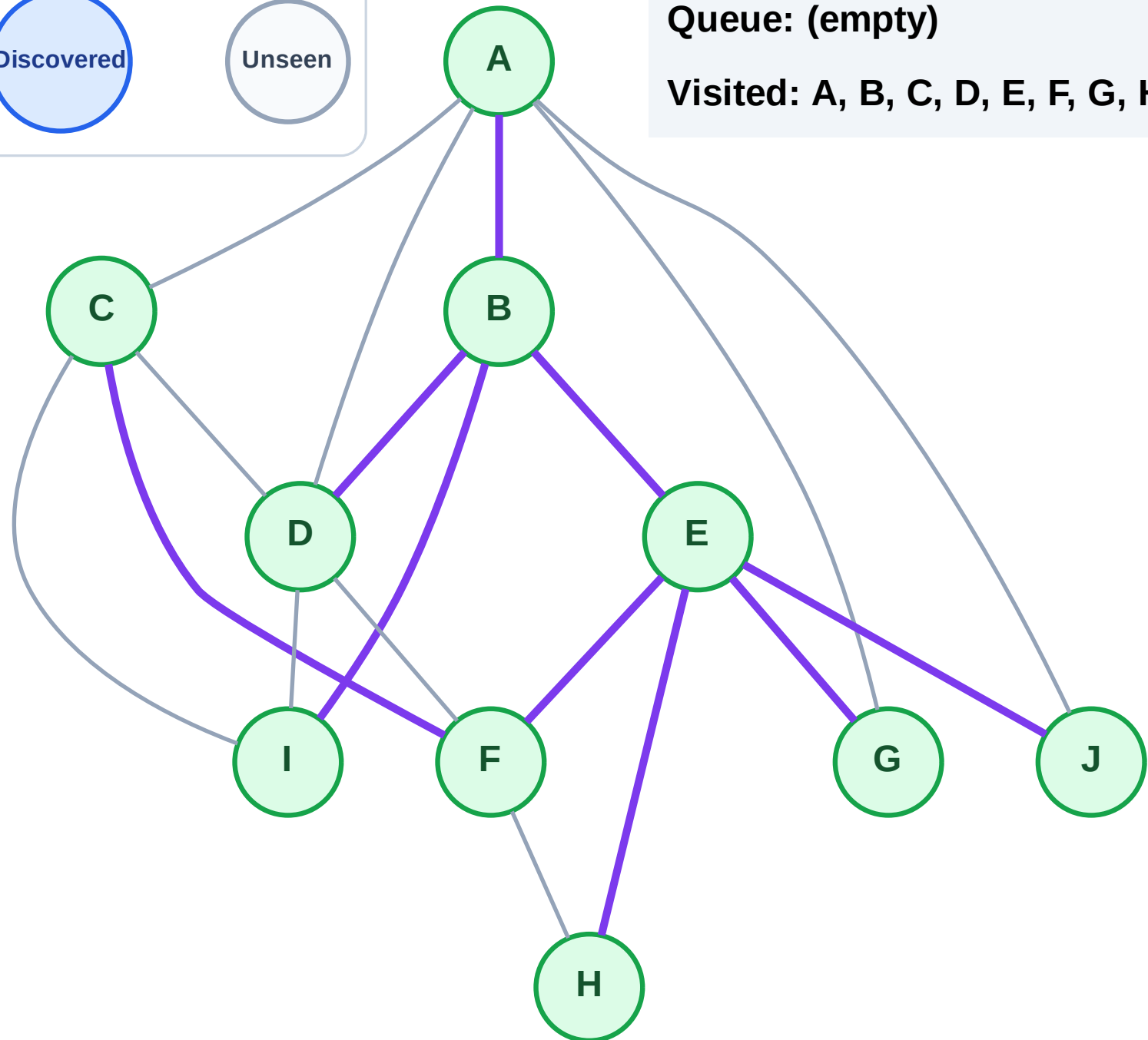
Step 21: No new neighbors from C

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

Complete

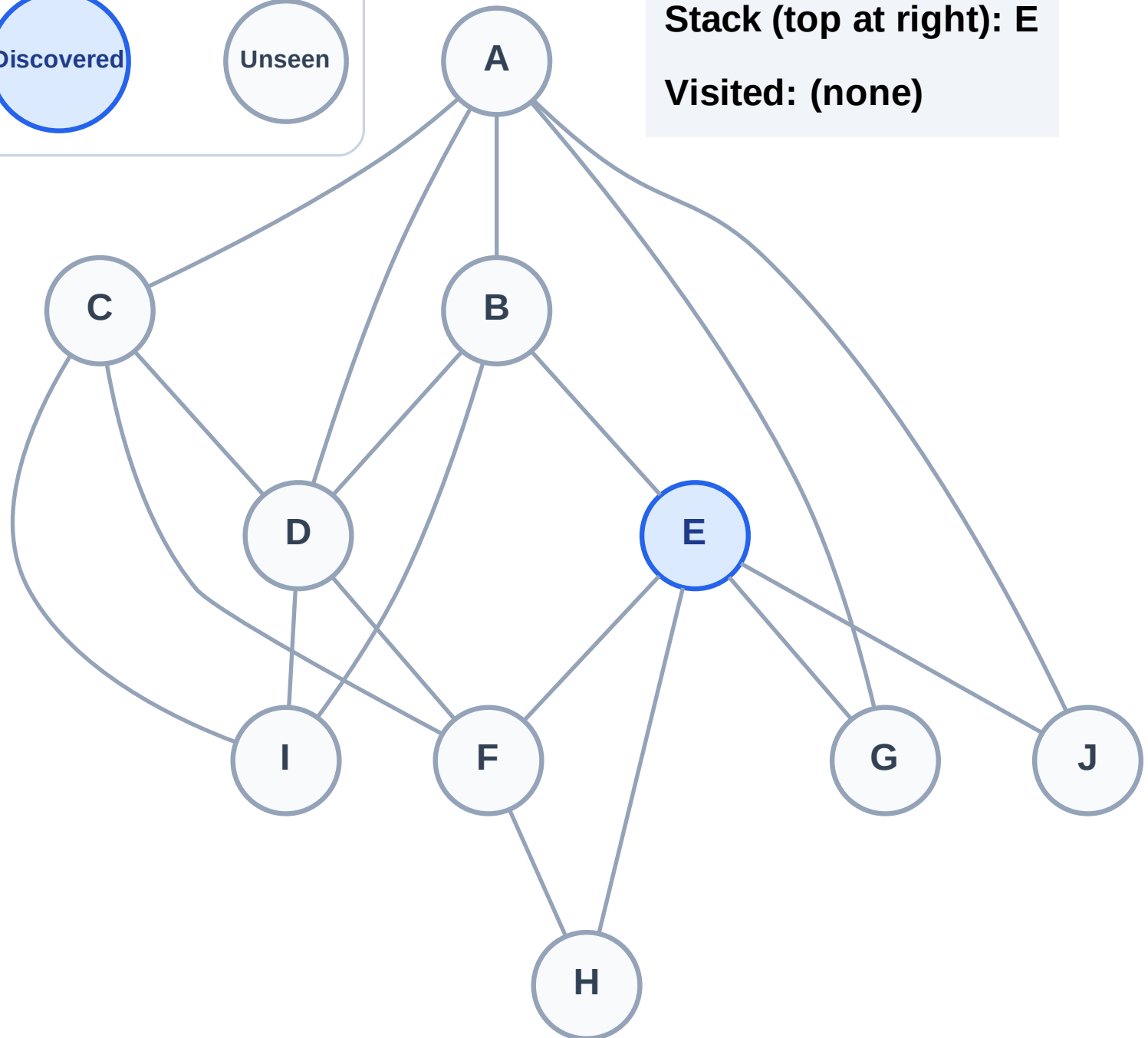
Processing

Discovered

Unseen

Stack (top at right): E

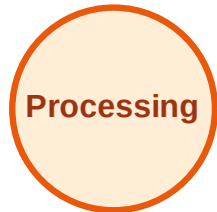
Visited: (none)



DFS Traversal

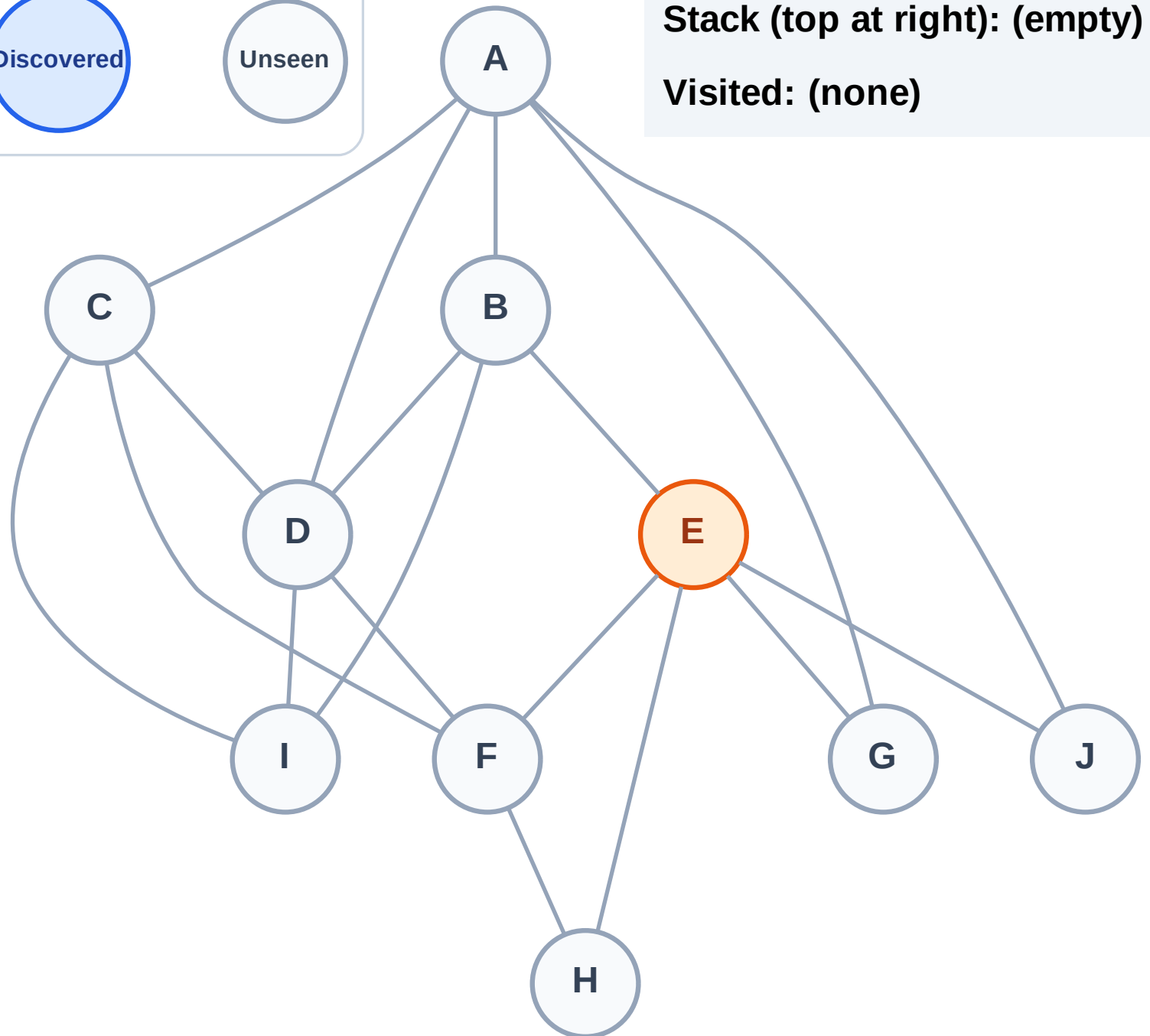
Step 2: Process node E

Legend



Stack (top at right): (empty)

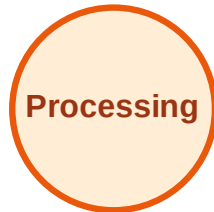
Visited: (none)



DFS Traversal

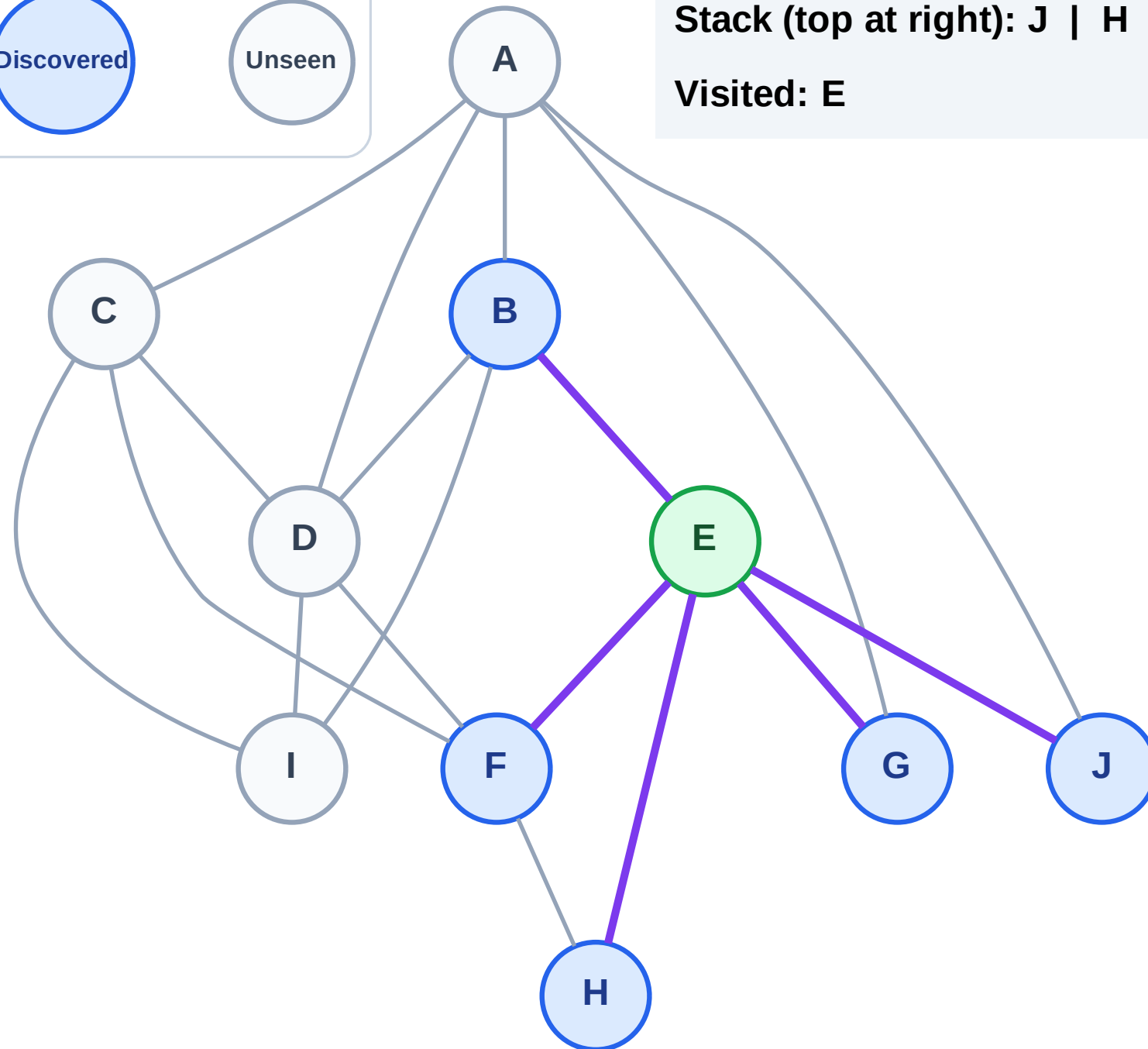
Step 3: Discover J, H, G, F, B from E

Legend



Stack (top at right): J | H | G | F | B

Visited: E



DFS Traversal

Step 4: Process node B

Legend

Complete

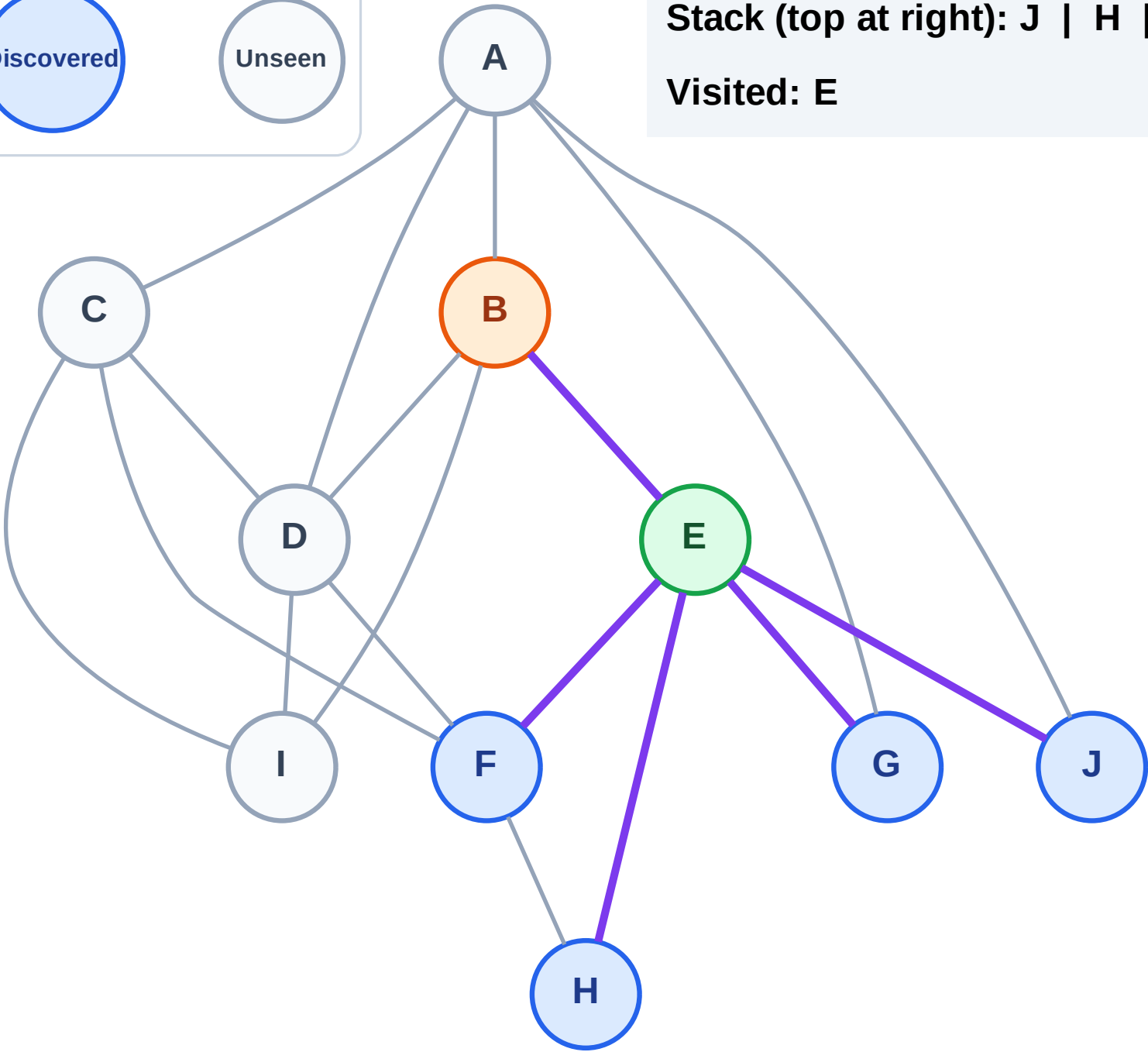
Processing

Discovered

Unseen

Stack (top at right): J | H | G | F

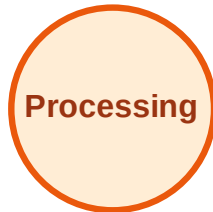
Visited: E



DFS Traversal

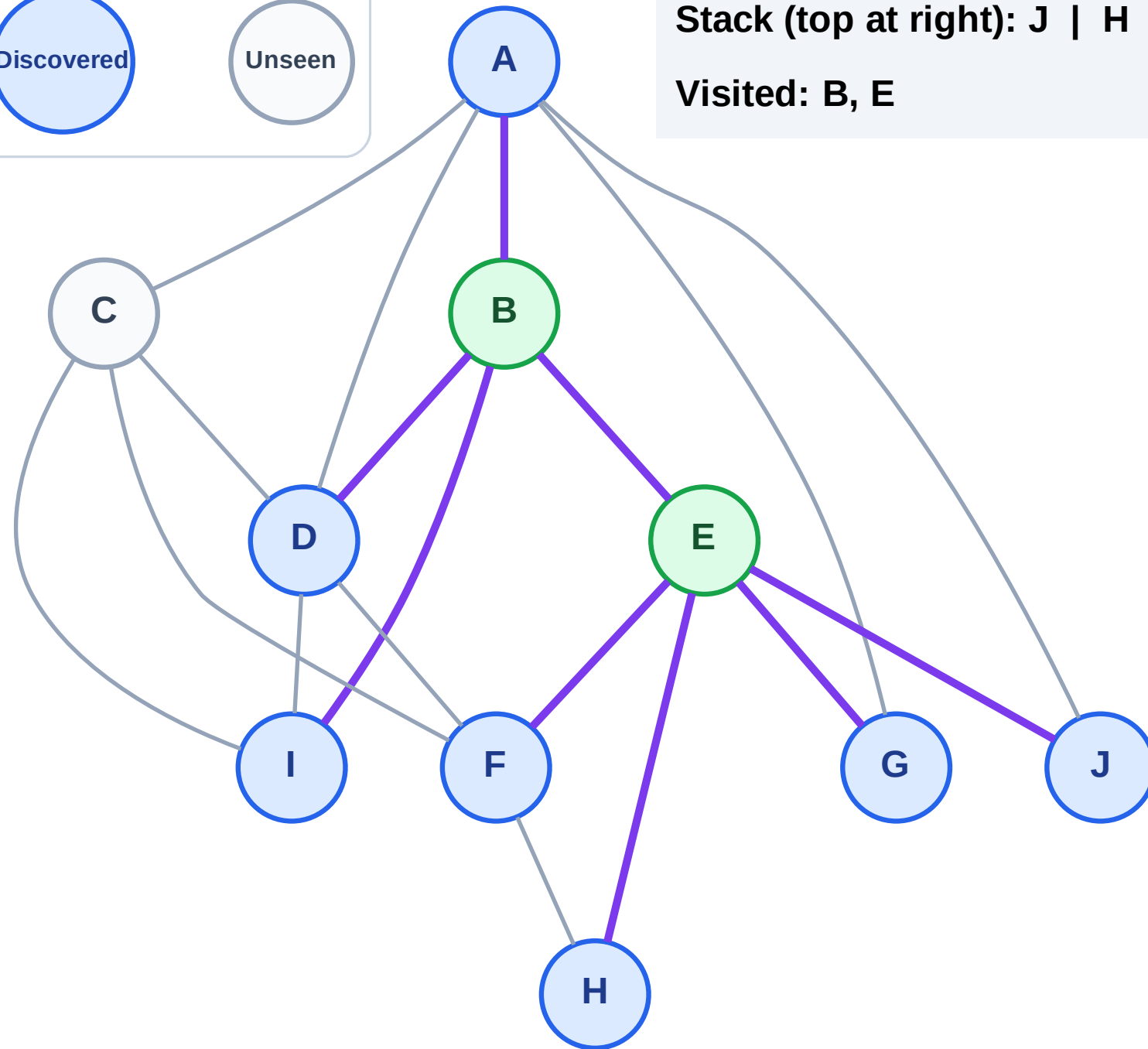
Step 5: Discover I, D, A from B

Legend



Stack (top at right): J | H | G | F | I | D | A

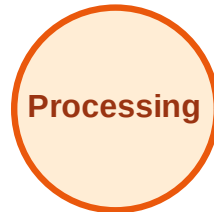
Visited: B, E



DFS Traversal

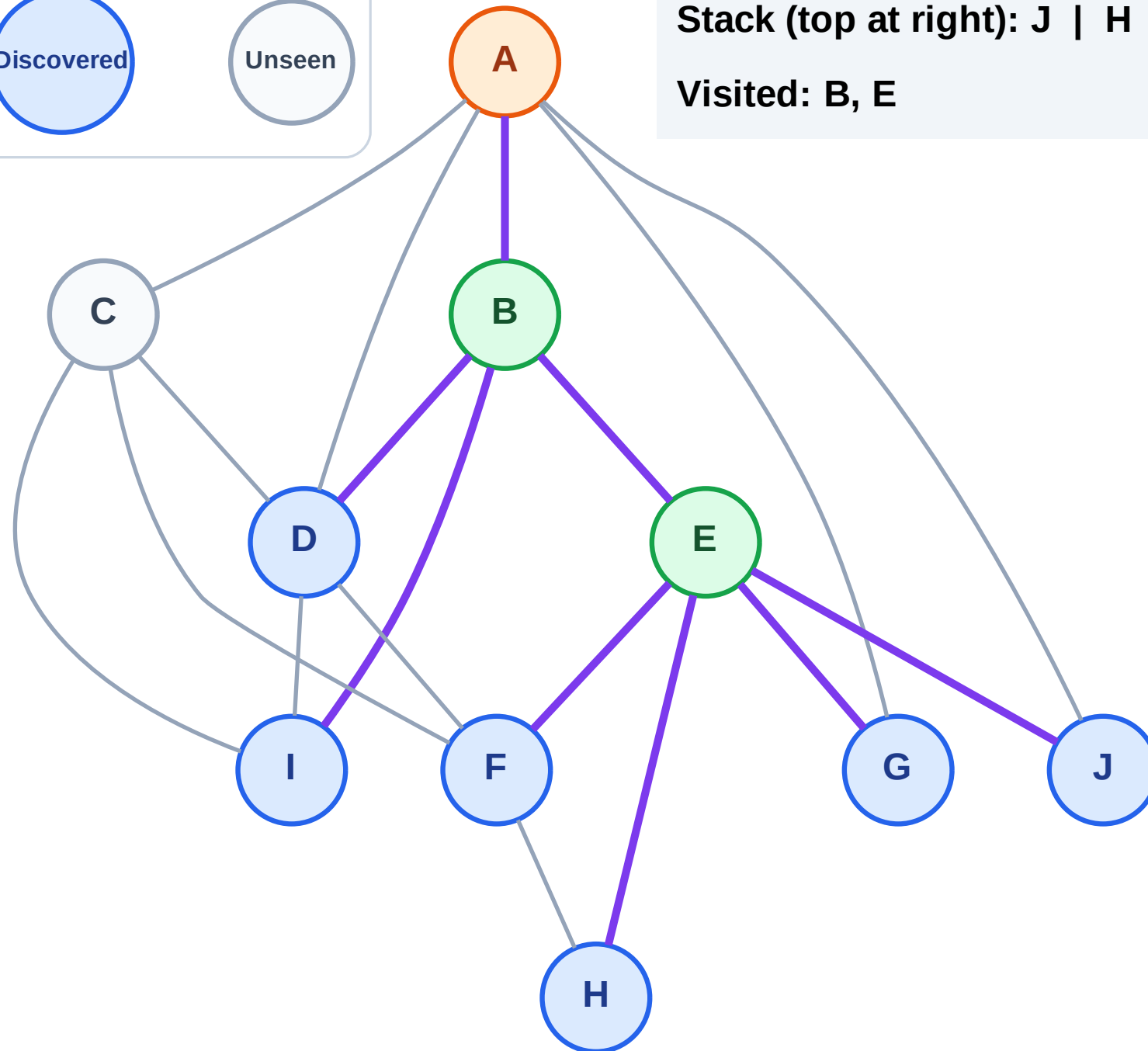
Step 6: Process node A

Legend



Stack (top at right): J | H | G | F | I | D

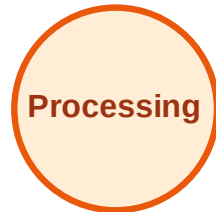
Visited: B, E



DFS Traversal

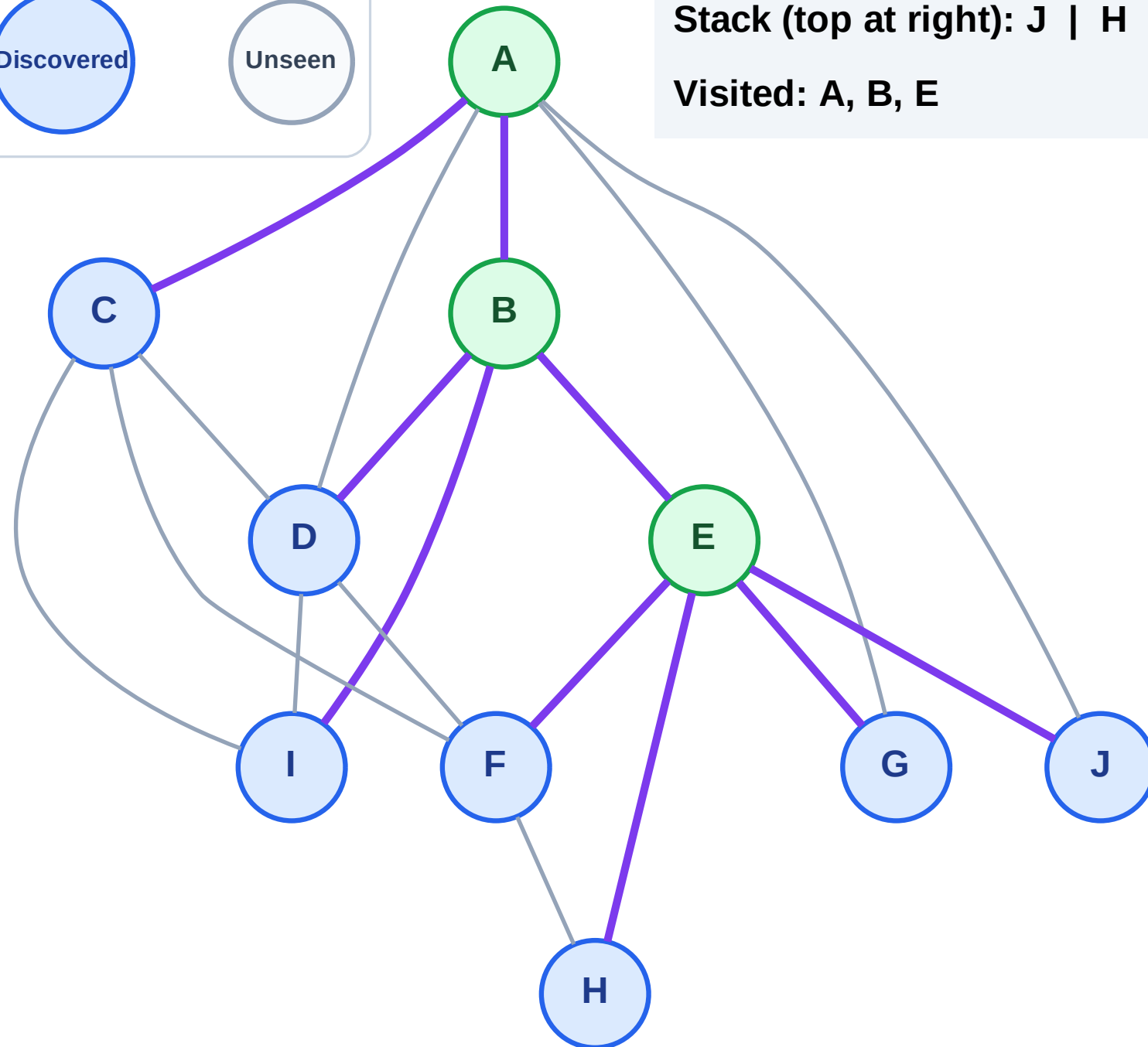
Step 7: Discover C from A

Legend



Stack (top at right): J | H | G | F | I | D | C

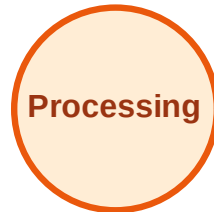
Visited: A, B, E



DFS Traversal

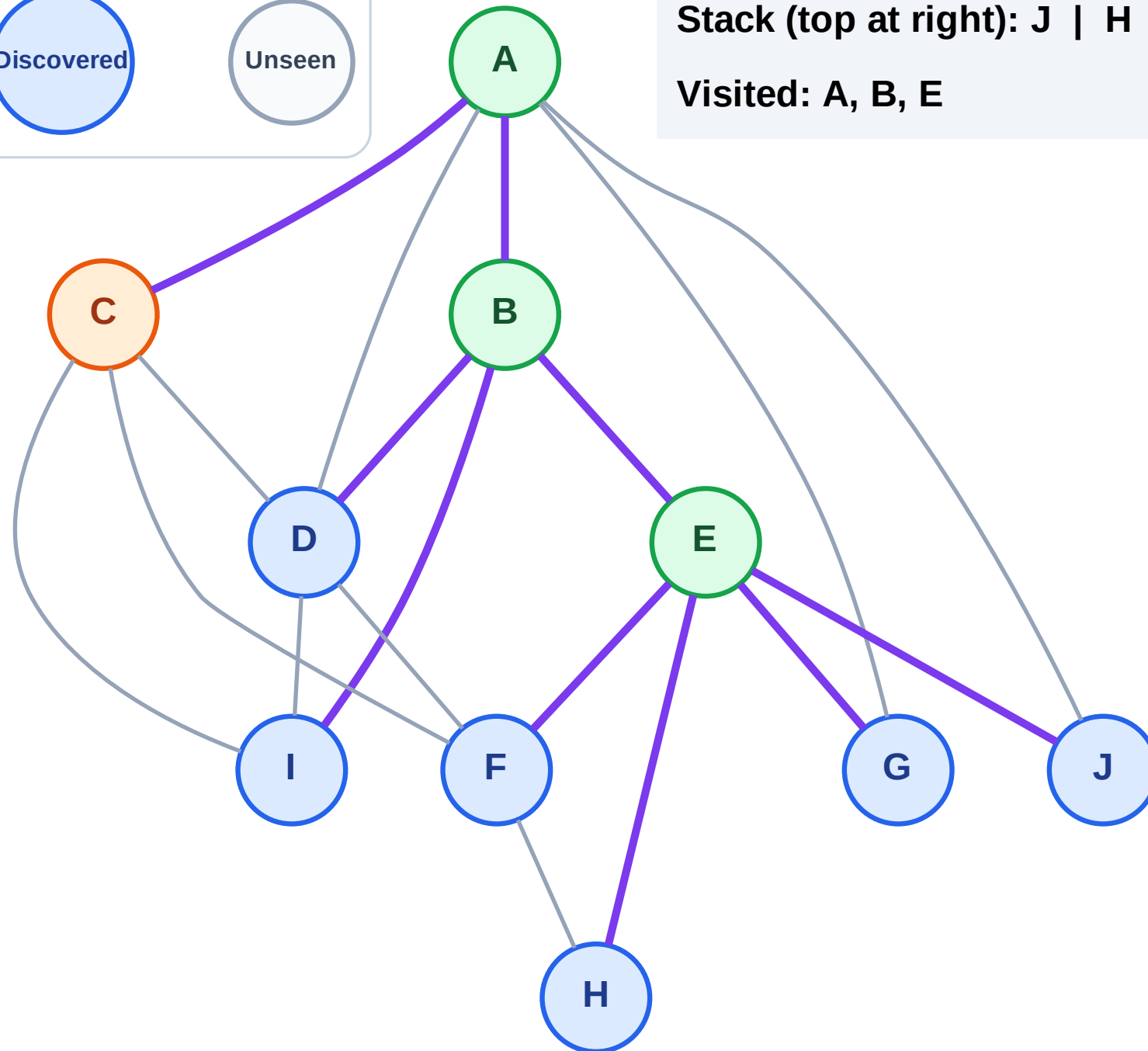
Step 8: Process node C

Legend



Stack (top at right): J | H | G | F | I | D

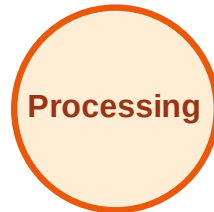
Visited: A, B, E



DFS Traversal

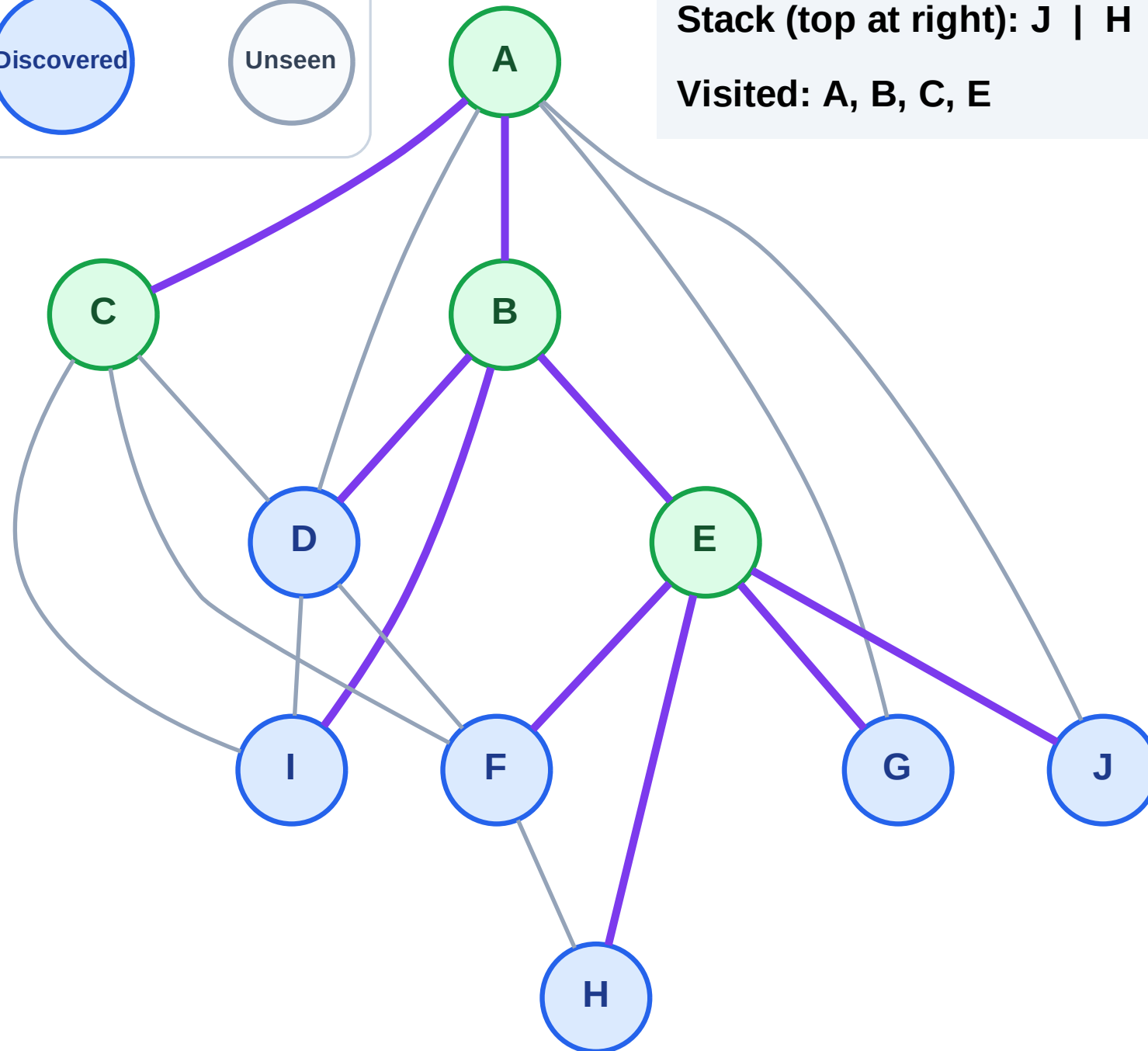
Step 9: No new neighbors from C

Legend



Stack (top at right): J | H | G | F | I | D

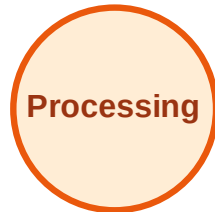
Visited: A, B, C, E



DFS Traversal

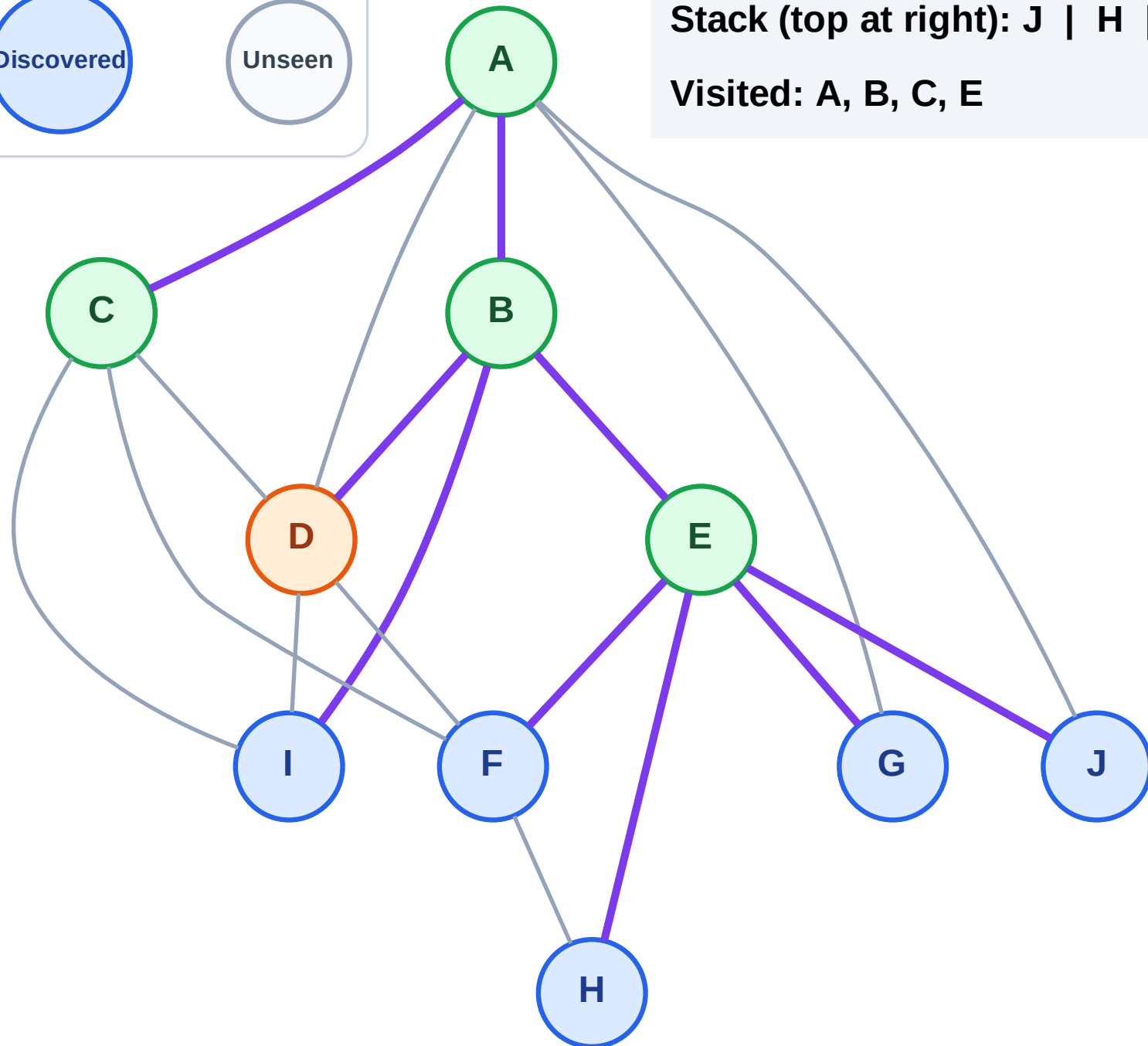
Step 10: Process node D

Legend



Stack (top at right): J | H | G | F | I

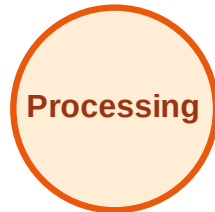
Visited: A, B, C, E



DFS Traversal

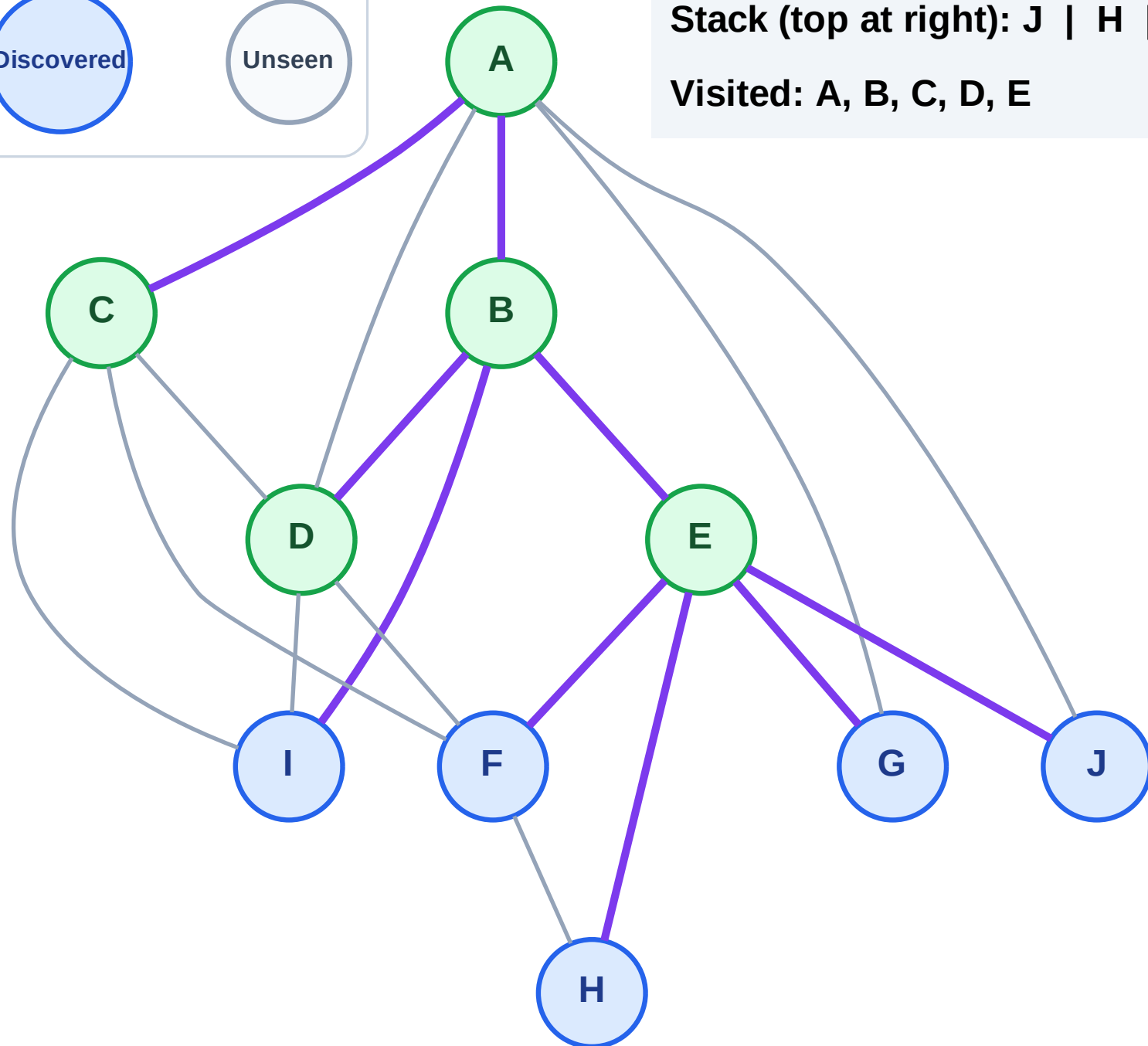
Step 11: No new neighbors from D

Legend



Stack (top at right): J | H | G | F | I

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node I

Legend

Complete

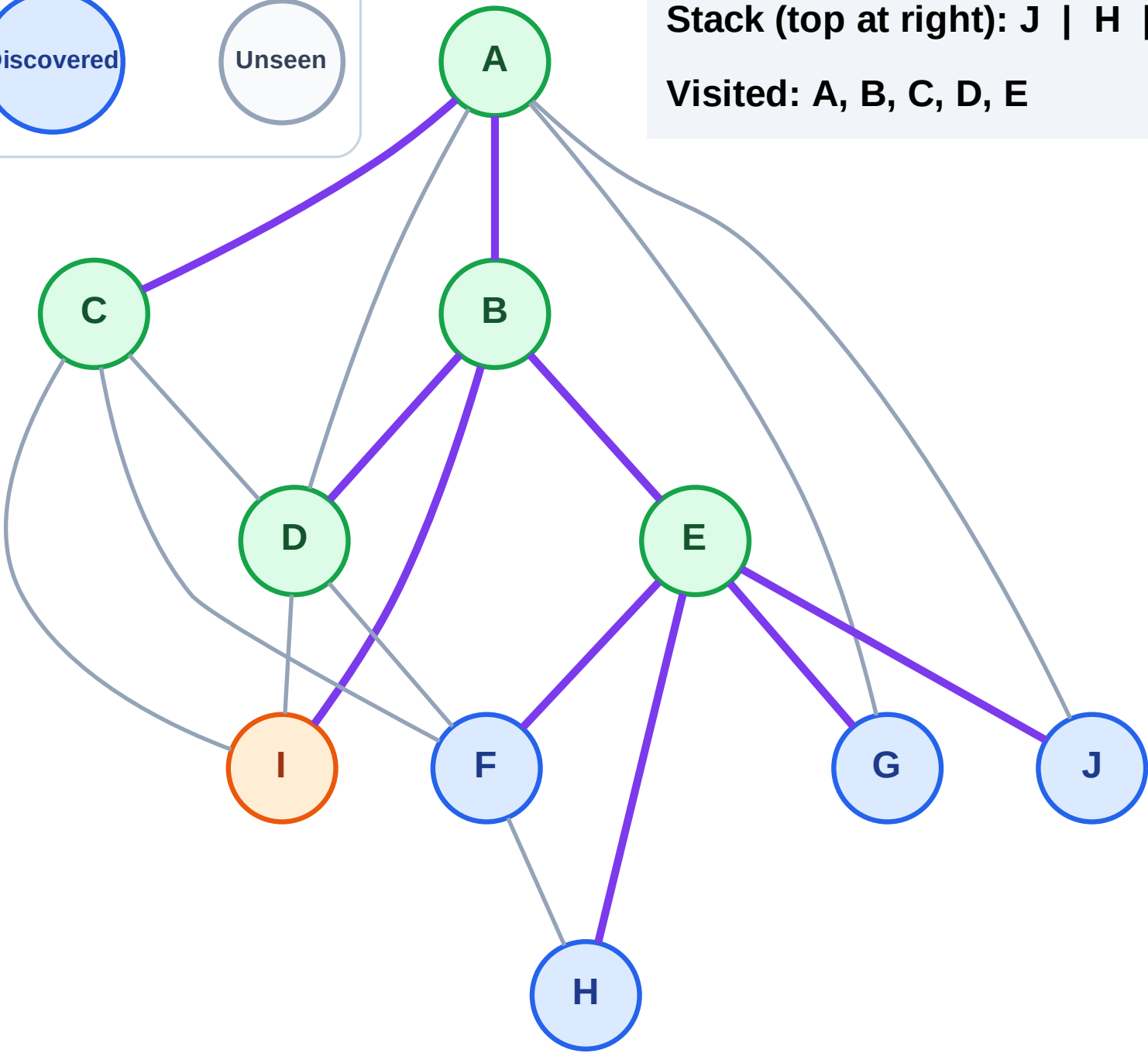
Processing

Discovered

Unseen

Stack (top at right): J | H | G | F

Visited: A, B, C, D, E



DFS Traversal

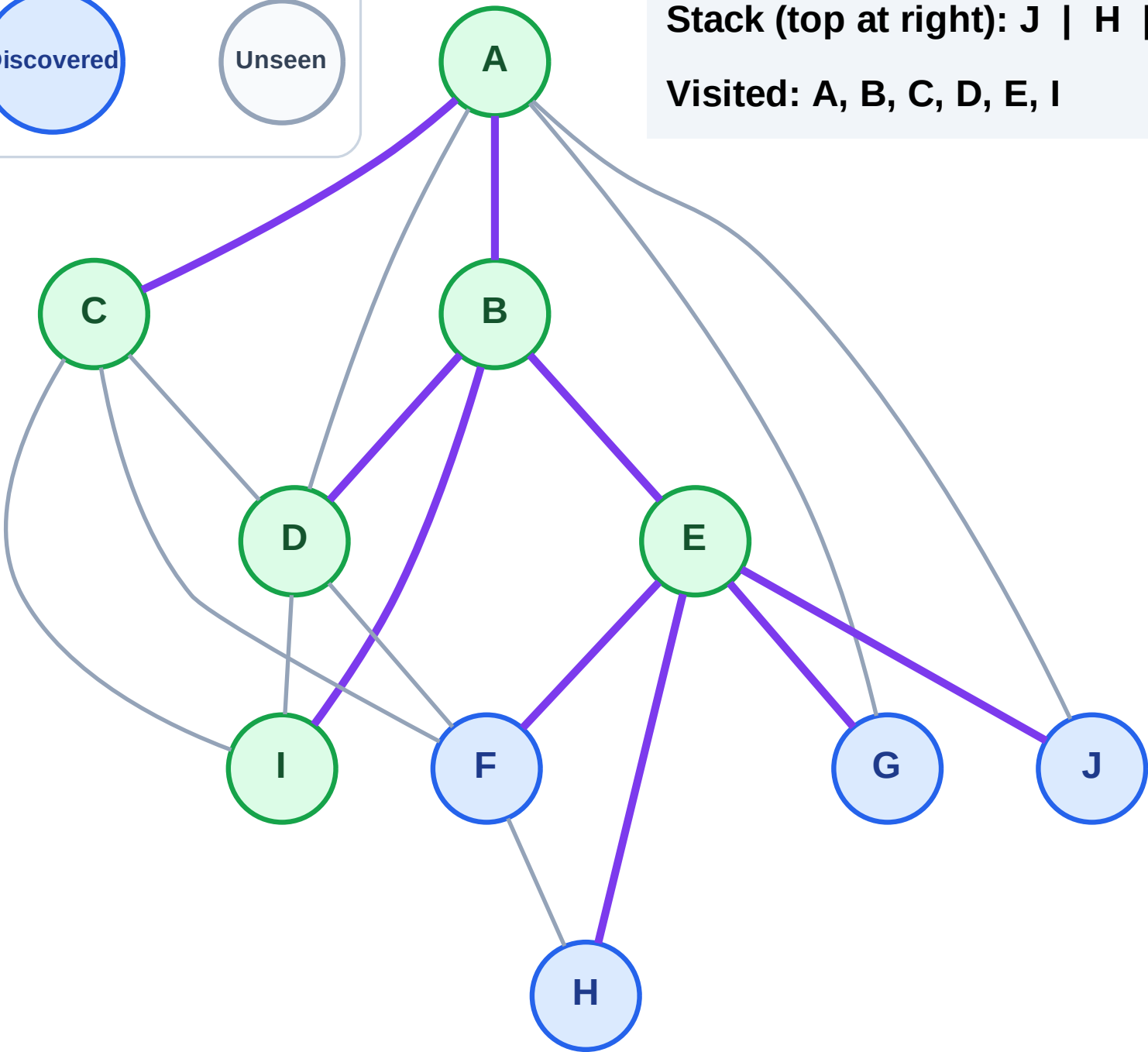
Step 13: No new neighbors from I

Legend



Stack (top at right): J | H | G | F

Visited: A, B, C, D, E, I



DFS Traversal

Step 14: Process node F

Legend

Complete

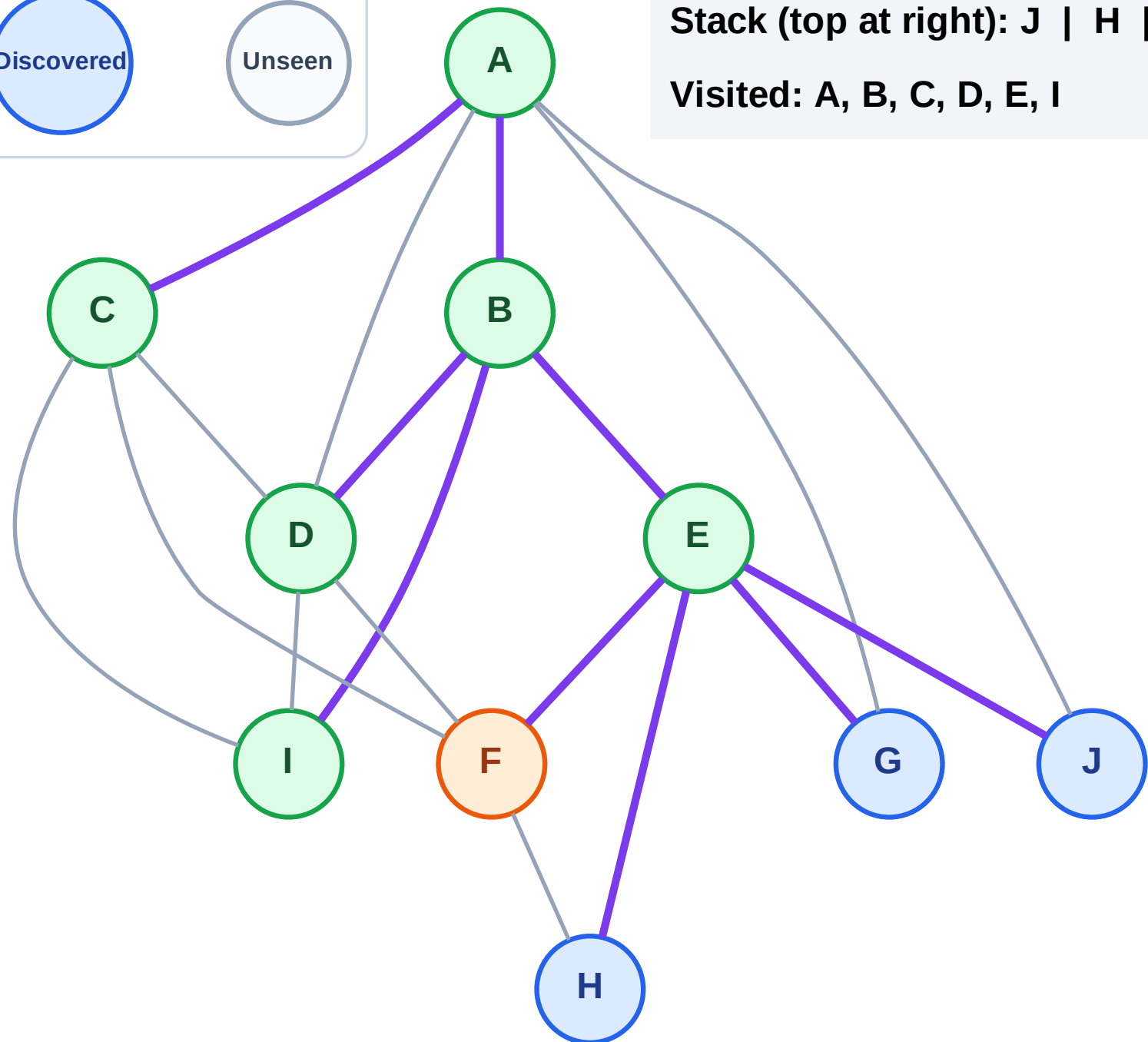
Processing

Discovered

Unseen

Stack (top at right): J | H | G

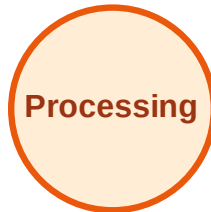
Visited: A, B, C, D, E, I



DFS Traversal

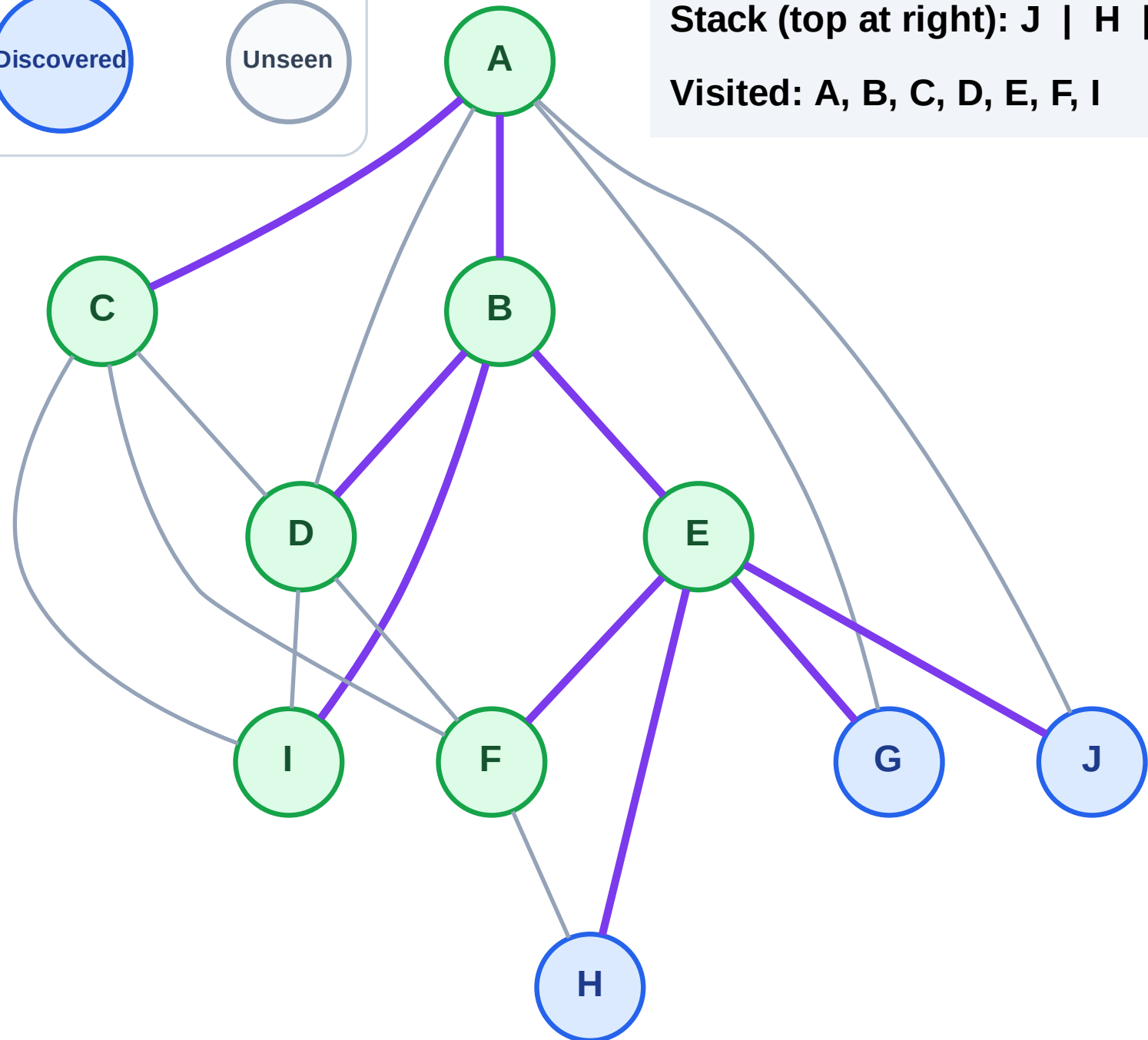
Step 15: No new neighbors from F

Legend



Stack (top at right): J | H | G

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 16: Process node G

Legend

Complete

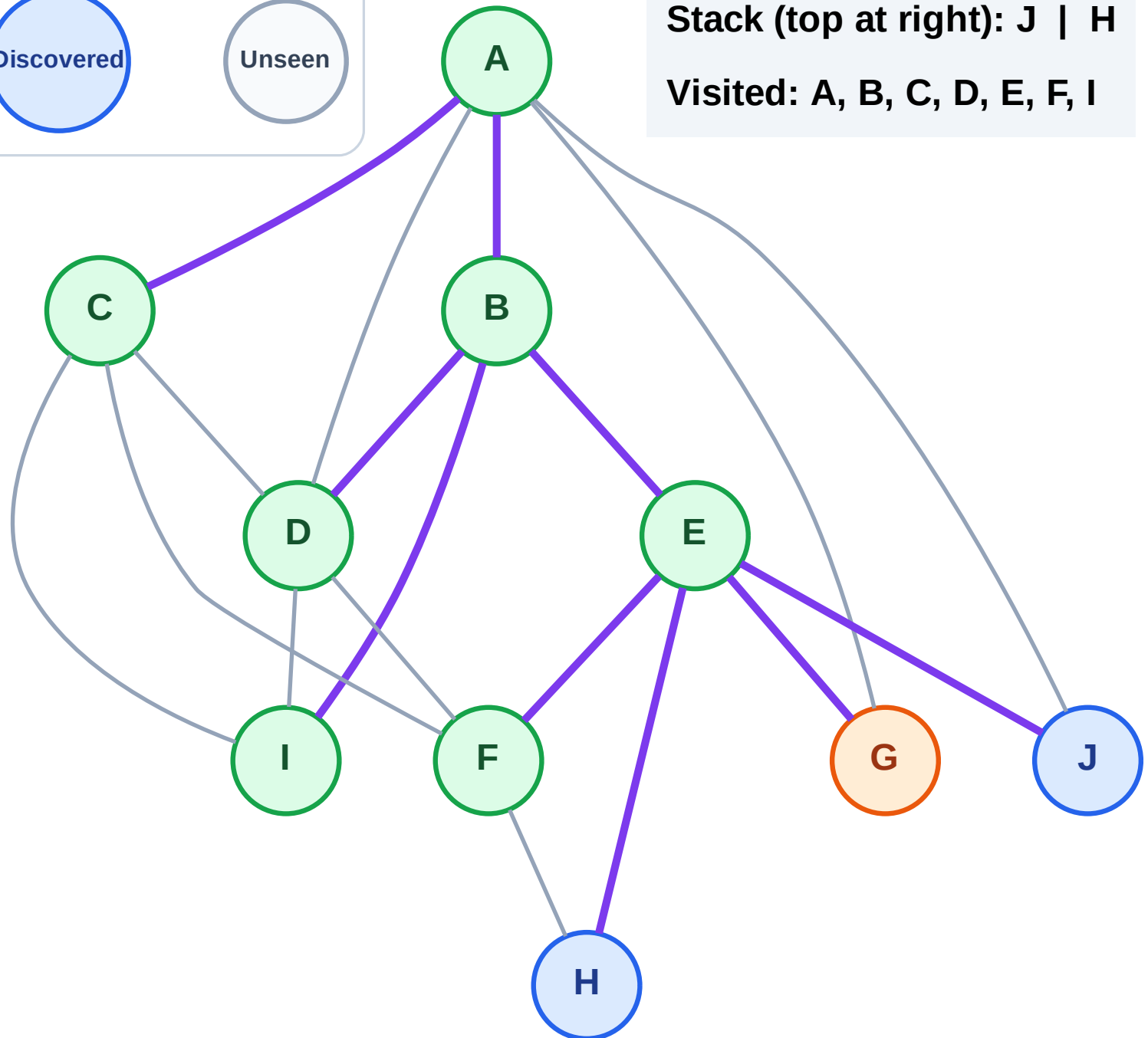
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 17: No new neighbors from G

Legend

Complete

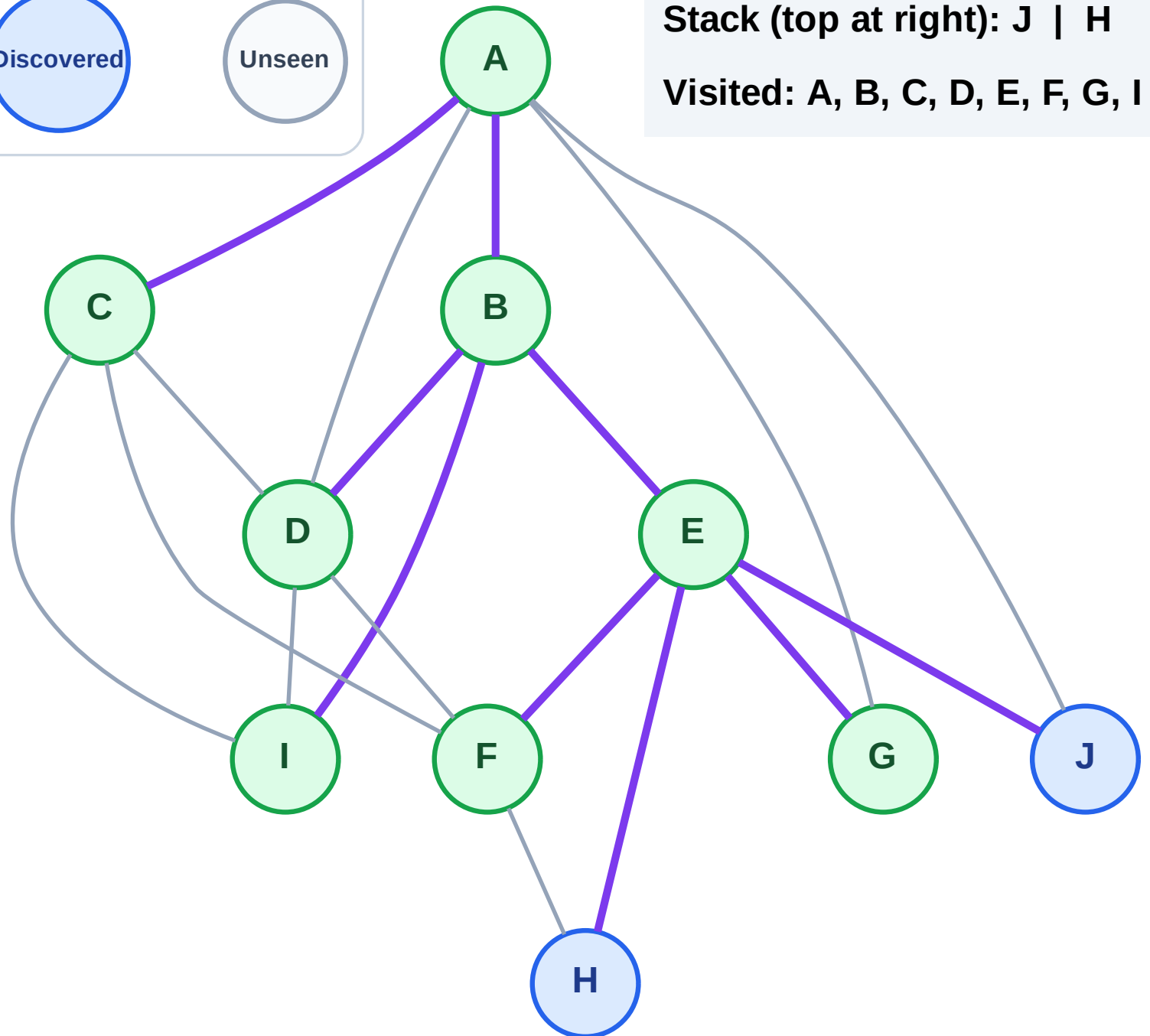
Processing

Discovered

Unseen

Stack (top at right): J | H

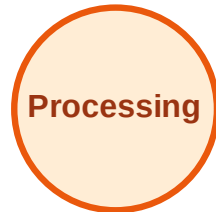
Visited: A, B, C, D, E, F, G, I



DFS Traversal

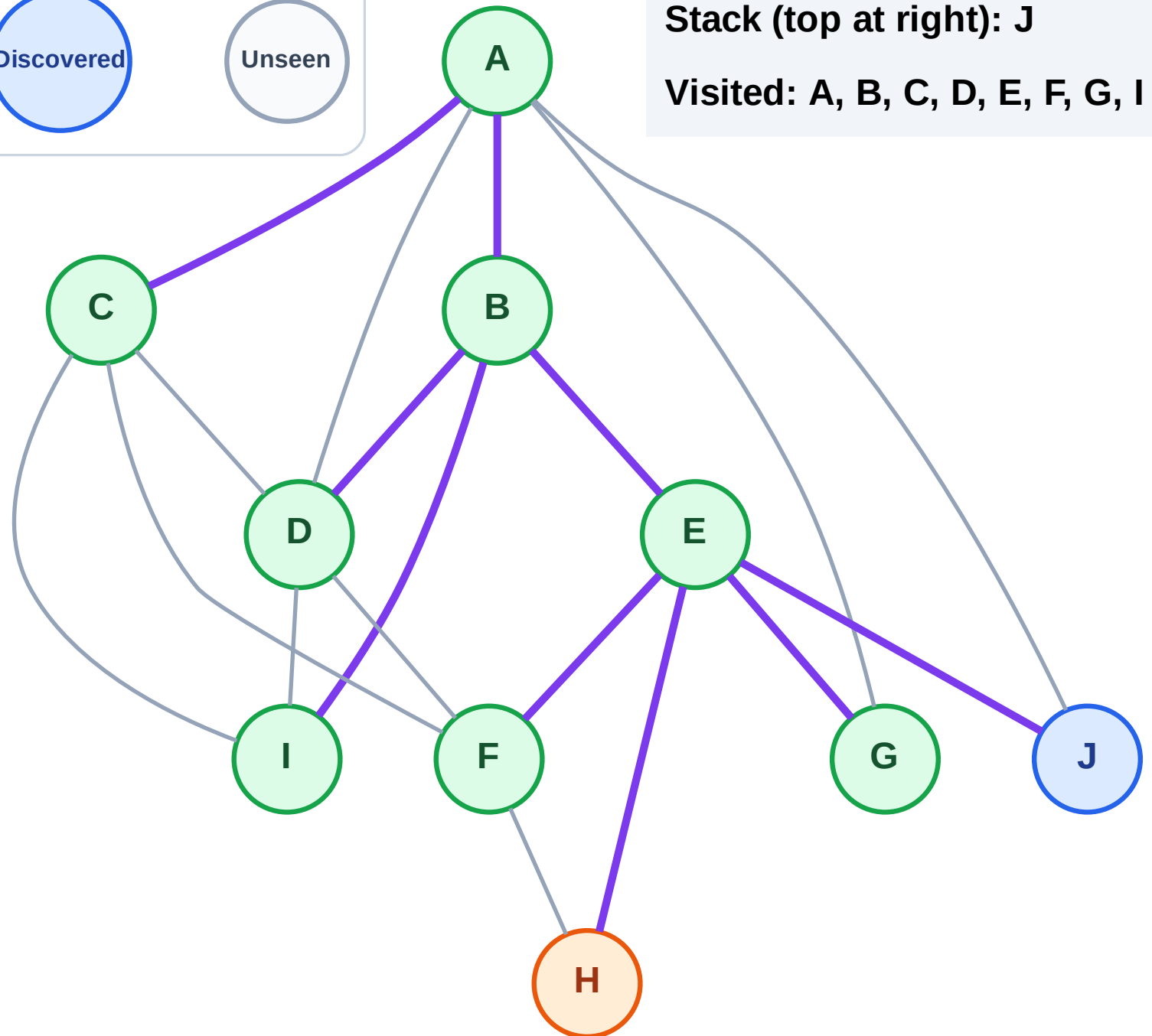
Step 18: Process node H

Legend



Stack (top at right): J

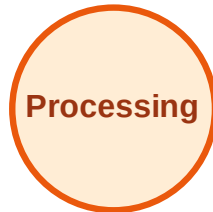
Visited: A, B, C, D, E, F, G, I



DFS Traversal

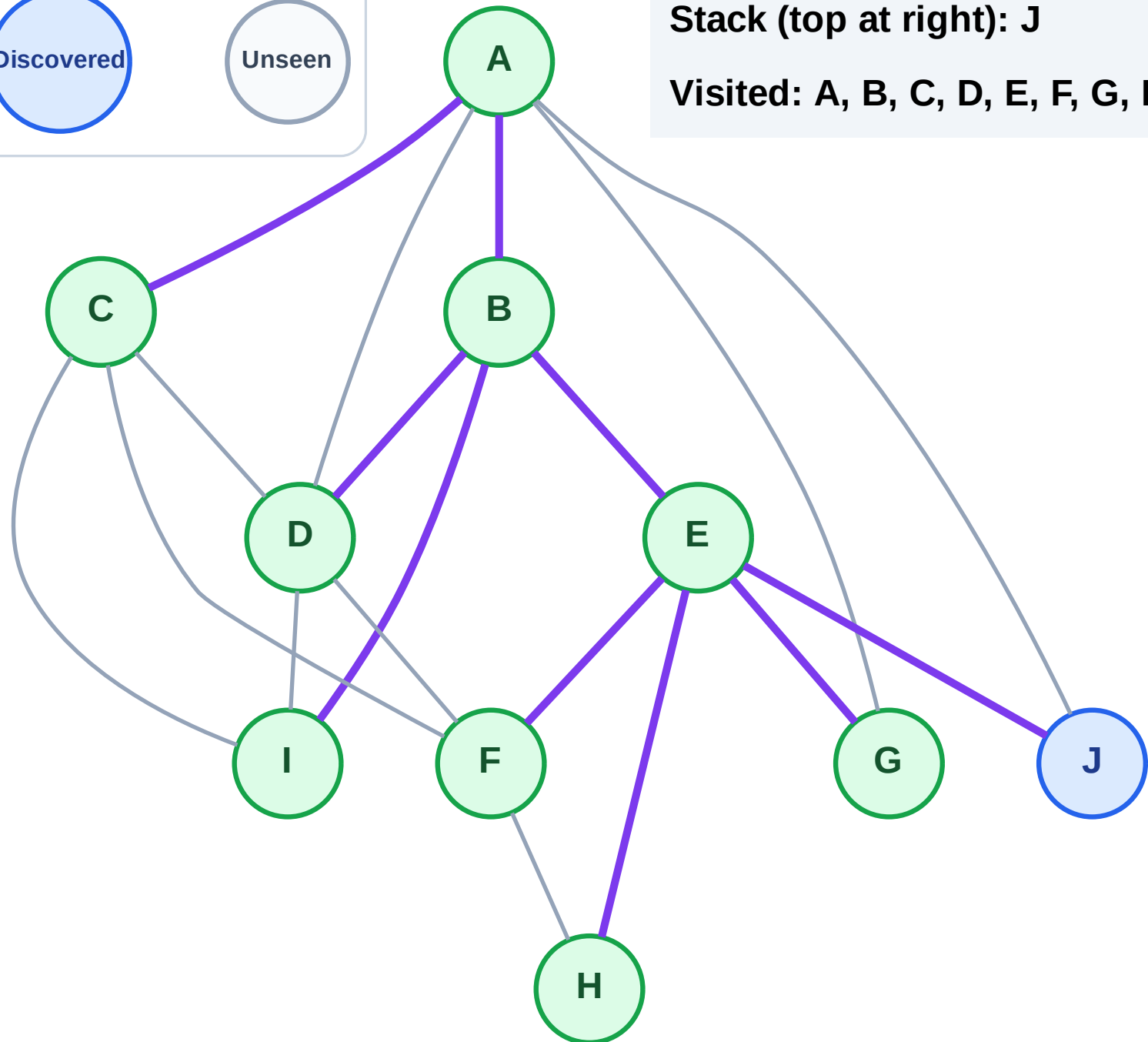
Step 19: No new neighbors from H

Legend



Stack (top at right): J

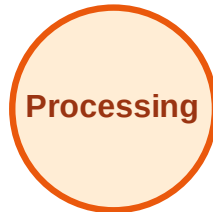
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

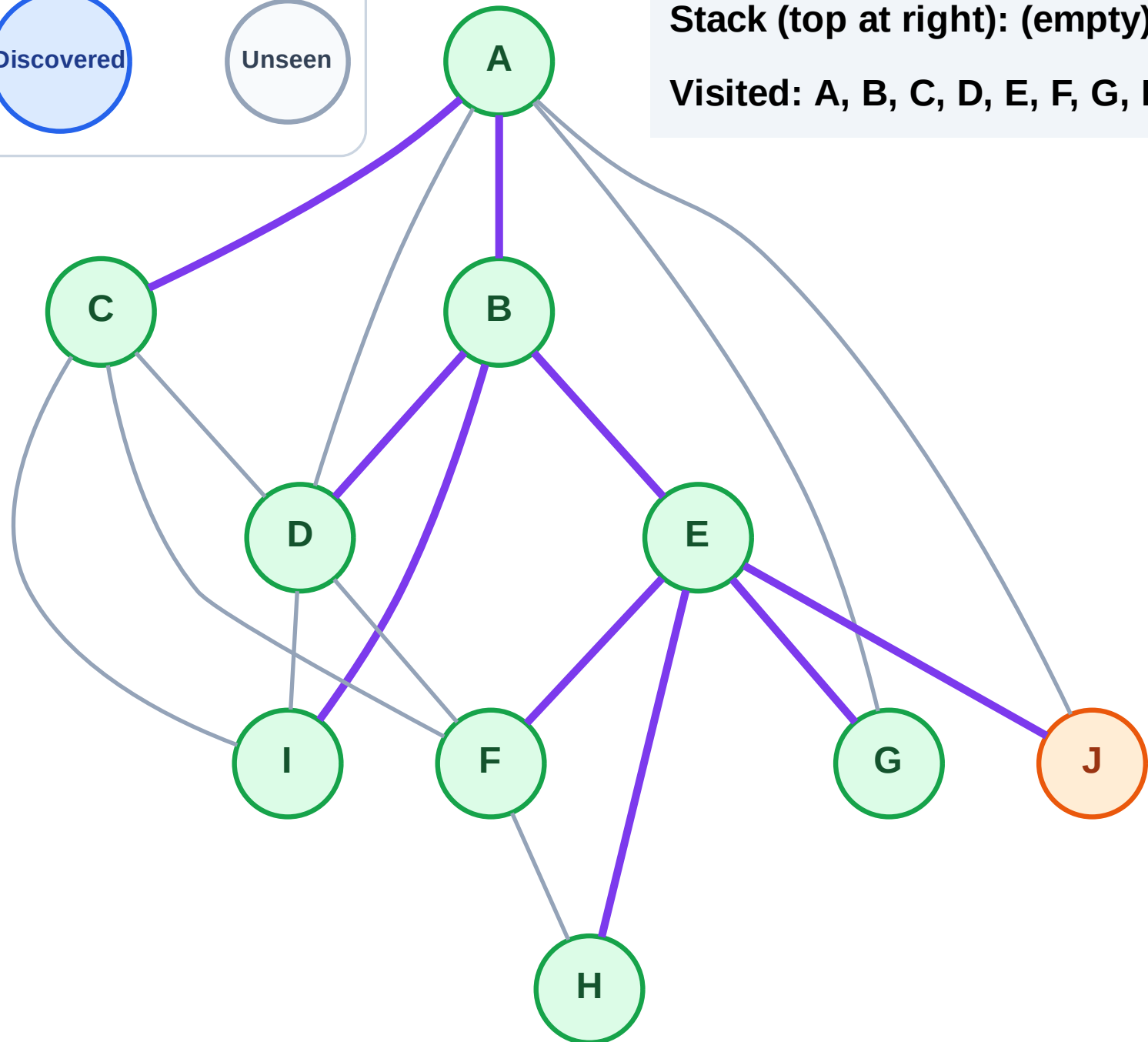
Step 20: Process node J

Legend



Stack (top at right): (empty)

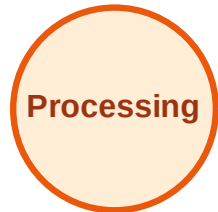
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

